

Custom Exam Paper

- 1** The universal set U consists of integers from 1 to 12 inclusive.
The sets A and B are defined within U as follows:

$$A = \{x \mid x \text{ is a multiple of } 3\}$$

$$B = \{x \mid x \text{ is a factor of } 12\}$$

- (a)** List the elements of the set $A \cap B$.

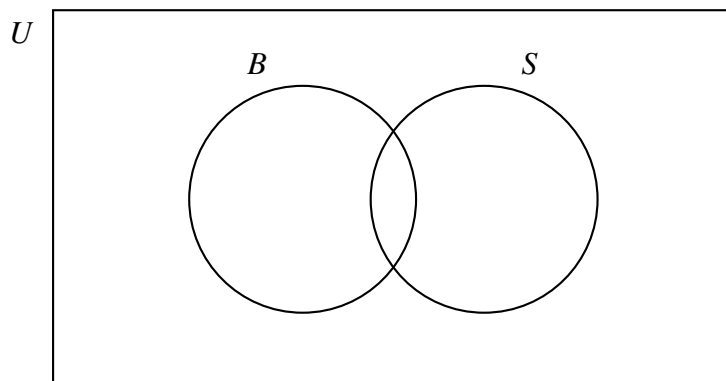
..... [2]

- (b)** Find $n(A \cup B')$.

..... [2]

- 2** A survey is conducted among a group of 30 students to find out who owns a bicycle (B) or a skateboard (S).
It is given that $n(U) = 30$, $n(B) = 17$, $n(S) = 13$, and $n(B \cap S) = 6$.

- (a)** Complete the Venn diagram to show the number of students in each region.



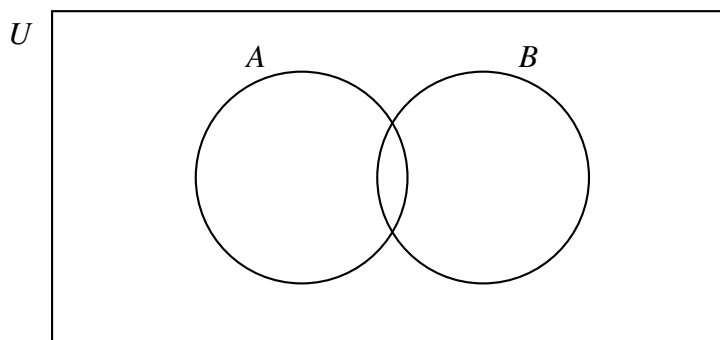
[3]

- (b) Find the number of students who do not own a bicycle.

..... [1]

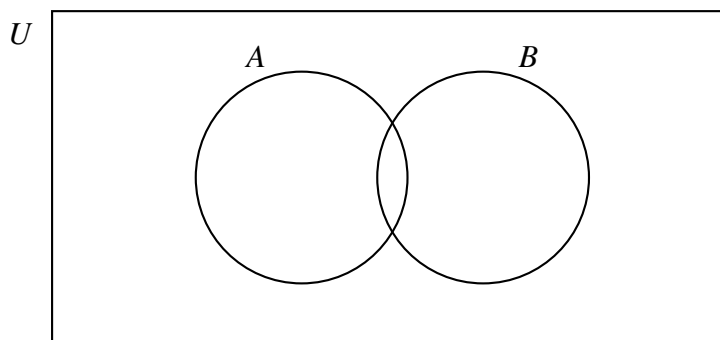
- 3 Shade the required regions on the copies of the Venn diagrams provided below.

- (a) On the diagram, shade the region representing $A' \cap B$.



[1]

- (b) On the diagram, shade the region representing $(A \cup B)'$.



[1]

- 4 The universal set U is defined as $\{x \mid x \text{ is an integer and } 1 \leq x \leq 10\}$.
The sets P and Q are subsets of U such that:

$$P = \{x \mid x \text{ is an even integer}\}$$

$$Q = \{x \mid x \text{ is a prime number}\}$$

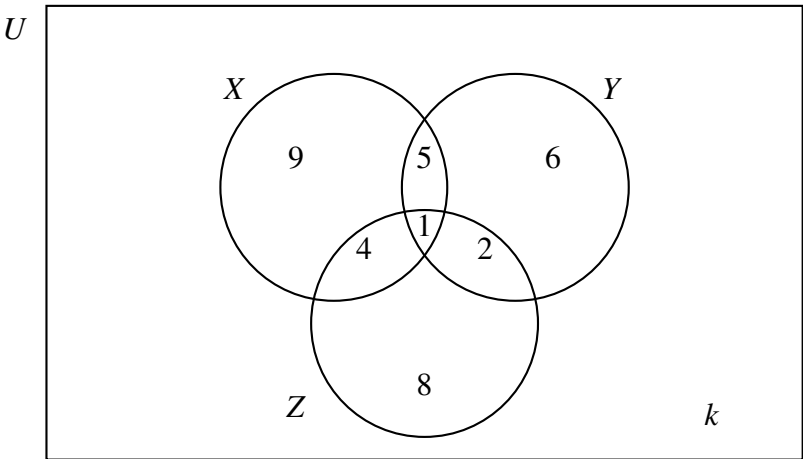
- (a) Use a standard set notation symbol to complete the statement about the number 2 and the set P .
- 2..... P

..... [1]

- (b) List all the elements of the set $(P \cup Q)'$.

..... [2]

- 5 The Venn diagram shows the number of elements in each region for the sets X , Y , and Z . It is given that $n(U) = 45$.



- (a) Calculate the value of k .

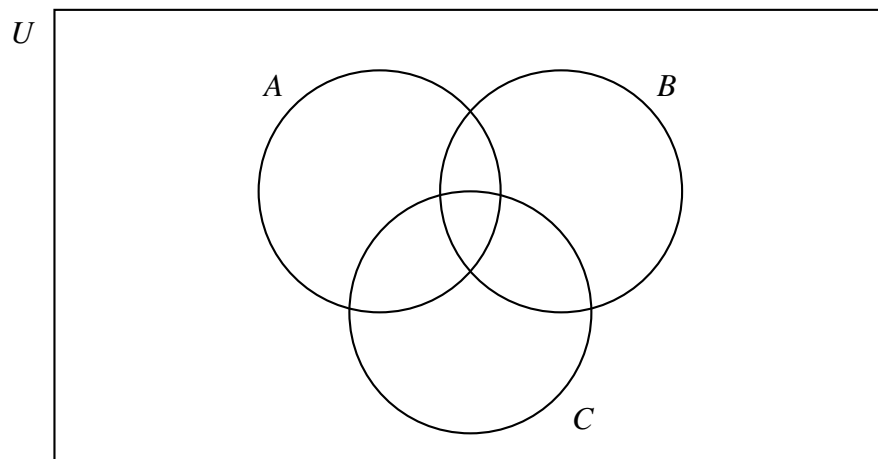
..... [2]

(b) Find $n(X \cap Y \cap Z')$.

..... [1]

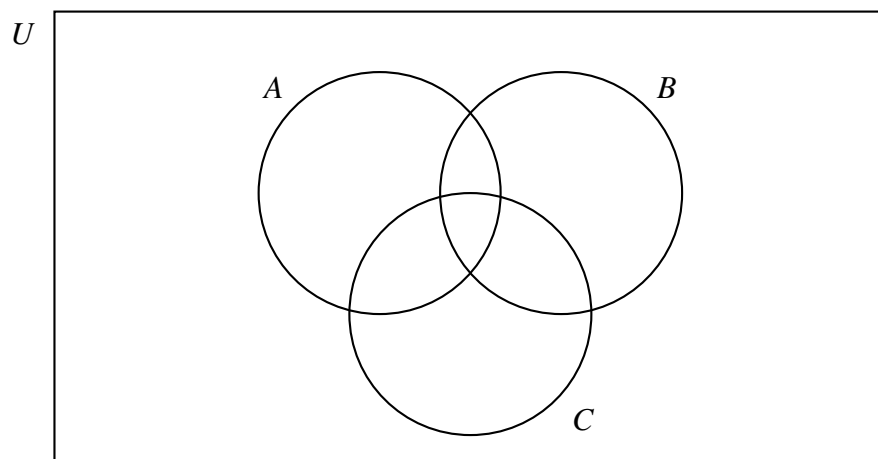
6 Shade the required regions on the copies of the Venn diagrams provided below.

(a) On the diagram, shade the region representing $A \cap (B \cup C)'$.



[1]

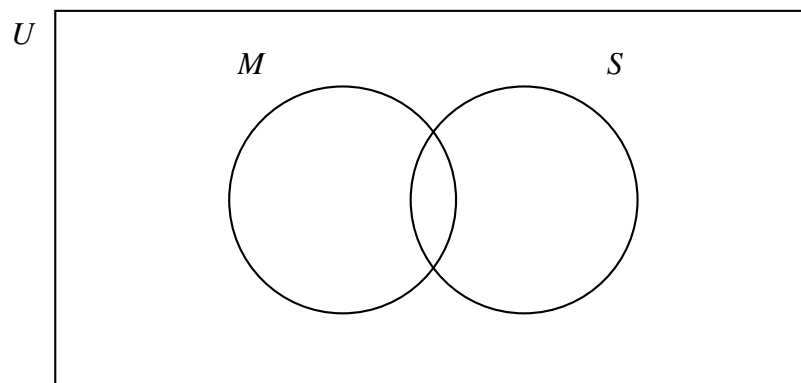
(b) On the diagram, shade the region representing $(A \cap B)' \cap C$.



[1]

- 7 A group of 40 students were asked if they like Mint chocolate (M) or Strawberry chocolate (S). It is given that $\frac{1}{4}$ of the total number of students like both flavours, 6 students like neither flavour, and the number of students who like Mint only is twice the number of students who like Strawberry only.

(a) Complete the Venn diagram to show the number of students in each region.



[3]

(b) Find $n(M \cup S')$.

..... [2]

- 8 The universal set U is defined as $\{x \mid x \text{ is a positive integer and } x \leq 20\}$. The sets A , B , and C are subsets of U defined as follows:

$$A = \{x \mid x \text{ is a multiple of } 4\}$$

$$B = \{x \mid x \text{ is a multiple of } 8\}$$

$$C = \{x \mid x \text{ is a factor of } 20\}$$

(a) Complete the statement using the correct standard set notation symbol to describe the relationship between set B and set A .

$$B \dots\dots A$$

..... [1]

(b) List all the elements of the set $(A \cap C) \cup B$.

..... [2]

9 The universal set U has $n(U) = 78$. Two sets P and Q are defined within U as follows:

$$P = \{y \mid y \in U \text{ and } y \text{ studies Physics}\}$$

$$Q = \{y \mid y \in U \text{ and } y \text{ studies Chemistry}\}$$

The number of elements in each specific region of the Venn diagram is given in terms of a positive integer x :

$$n(P \cap Q) = x^2 - 2x$$

$$n(P \cap Q') = 6x$$

$$n(Q \cap P') = 2x^2 - 5$$

$$n(P' \cap Q') = 3x + 7$$

(a) Show that the equation for the total number of elements reduces to $3x^2 + 7x - 76 = 0$.

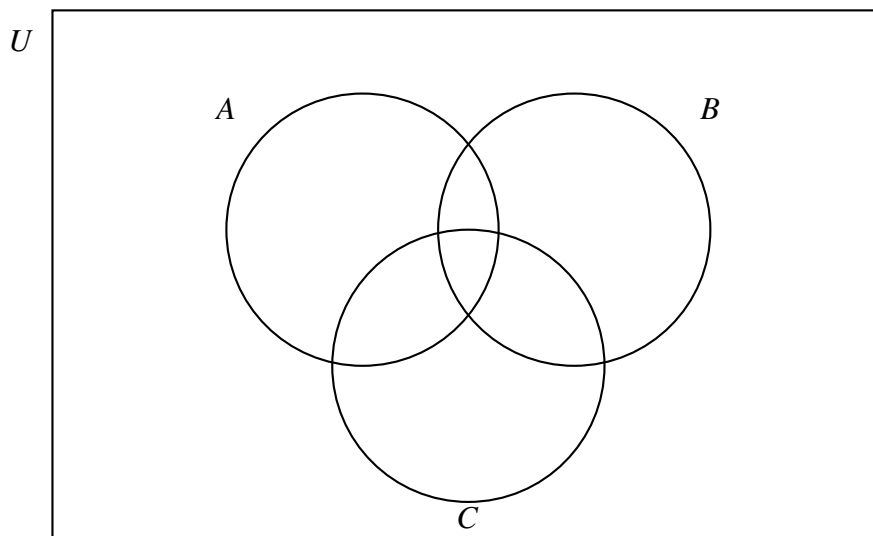
[2]

(b) Find $n(P' \cup Q)$.

..... [3]

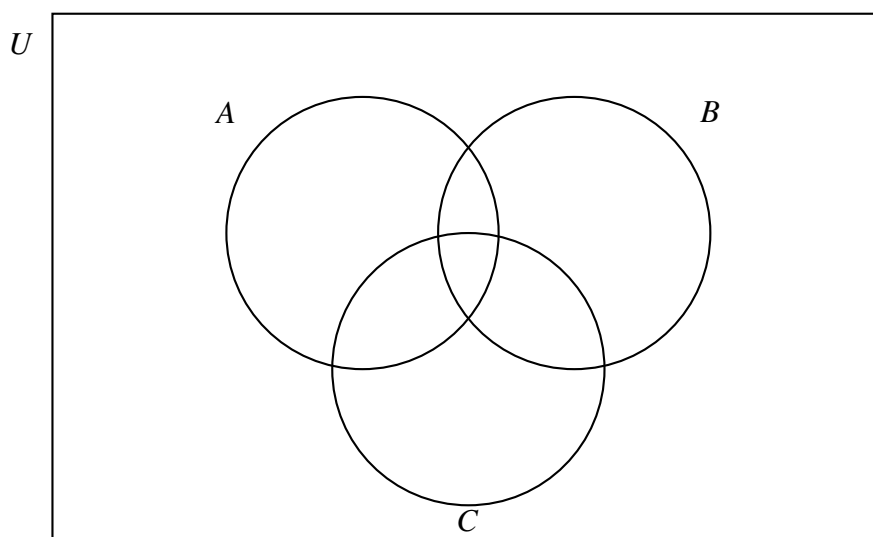
10 On the independent Venn diagrams provided below, shade the precise region specified by each set expression.

(a) Shade the region representing $(A \cap B') \cup (A' \cap B \cap C)$.



[2]

(b) Shade the region representing $(A \cup C)' \cap B$.



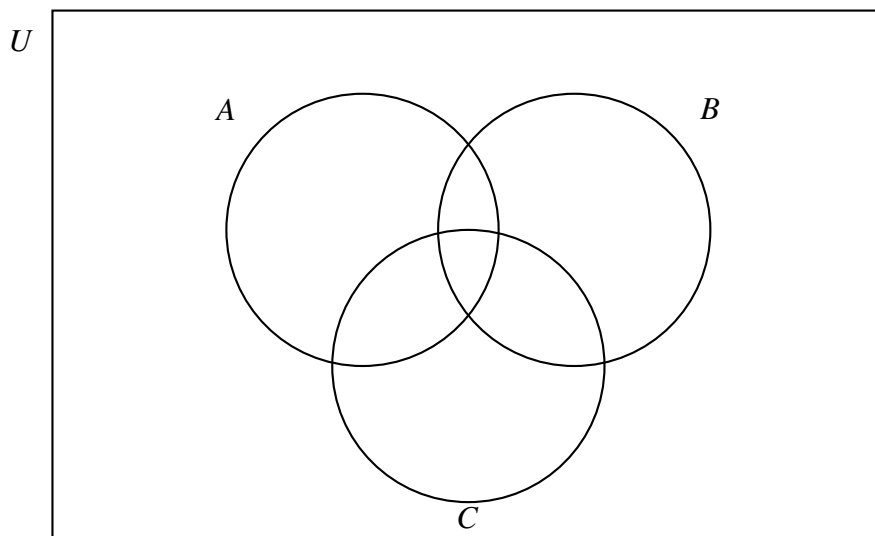
[2]

- 11** A group of 100 students are surveyed about their participation in Athletics (A), Badminton (B), and Cricket (C). The results show that 38 students do Athletics, 34 students do Badminton, and 41 students do Cricket. The number of students in the intersection regions are described in terms of x :
- * x students participate in all three sports.
 - * The number of students who do Badminton and Cricket only is $2x$.
 - * The number of students who do Athletics and Cricket only is $x + 1$.
 - * The number of students who do Athletics and Badminton only is $2x - 3$.
 - * The number of students who do Cricket only is equal to the number of students who do none of the three sports.

- (a) Formulate an equation in terms of x for the total number of students and show that it reduces to $155 - 11x = 100$.

[3]

- (b) Complete the Venn diagram to show the exact number of students in each of the 8 regions.



[3]

- 12** The universal set U is defined as:

$$U = \{x \mid x \text{ is an integer and } 1 \leq x \leq 30\}$$

Three subsets A , B , and C are defined within U as follows:

$$A = \{x \mid x \in U \text{ and } x \text{ is a multiple of } 3\}$$

$$B = \{x \mid x \in U \text{ and } x \text{ is a multiple of } 4\}$$

$$C = \{x \mid x \in U \text{ and } x \text{ is a perfect square}\}$$

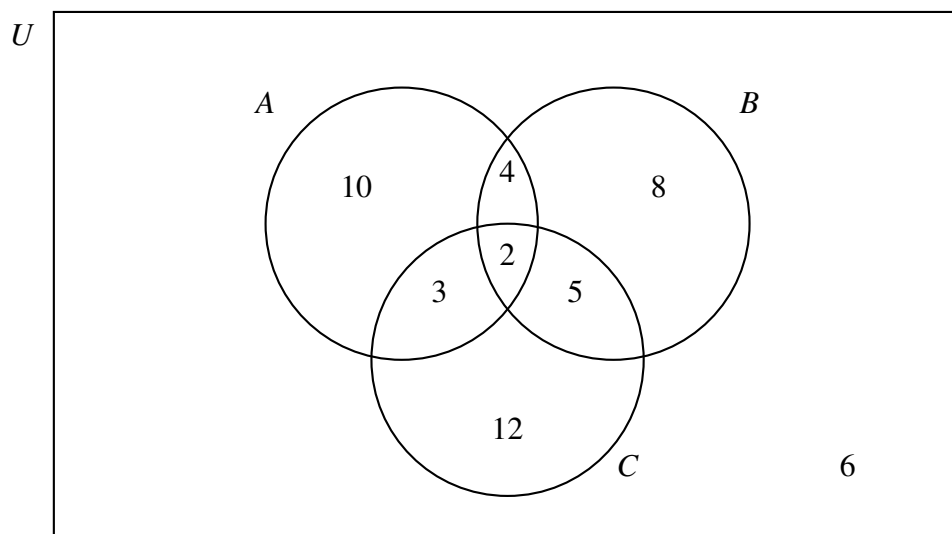
- (a) State whether the statement $C \subseteq (A' \cap B)'$ is true or false. Justify your answer explicitly by identifying the elements that support your conclusion.

[2]

- (b) Calculate $n((A \cap B') \cup (B \cap C'))$.

..... [2]

- 13 The Venn diagram shows the number of elements in three sets, A , B , and C .



(a) Find $n(A \cap B)$.

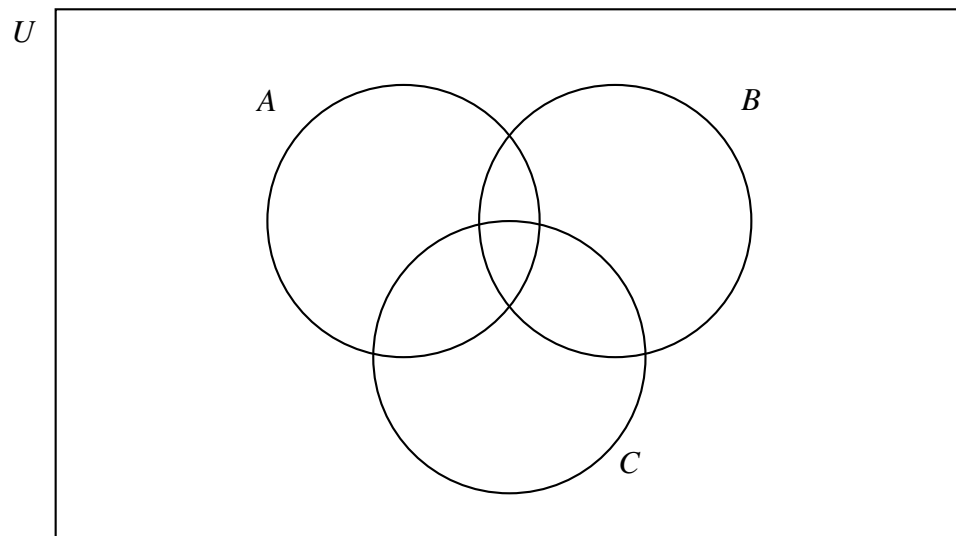
..... [1]

(b) Find $n(A \cup C)$.

..... [2]

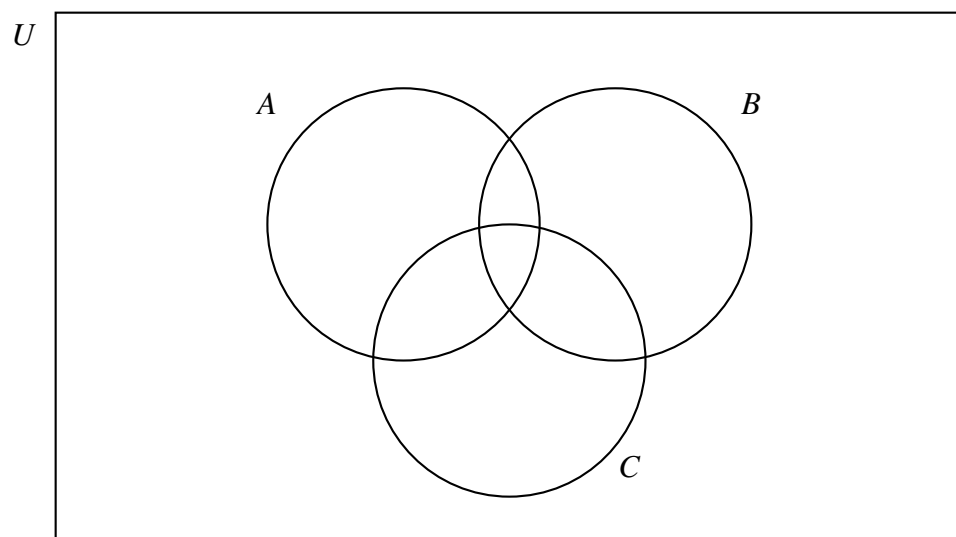
- 14** On the independent Venn diagrams provided below, shade the precise region specified by each set expression.

(a) On the diagram, shade the region representing $A \cap B \cap C'$.



[1]

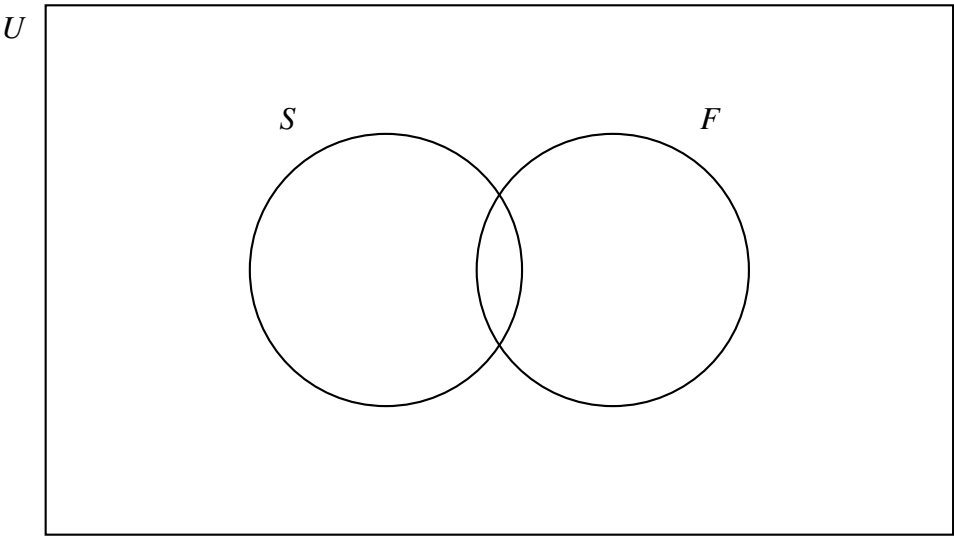
(b) On the diagram, shade the region representing $(A \cup B)' \cap C$.



[1]

15 A survey of 50 people found that 28 speak Spanish (S) and 25 speak French (F). There are 8 people who speak neither language.

(a) Complete the Venn diagram to show the number of people in each region.



[3]

(b) Find the number of people who speak Spanish but do not speak French.

..... [1]

16 **(a)** Write the number four million five hundred thousand and twenty-six in figures.

..... [1]

(b) From the following list of numbers:

$\sqrt{7}$, 9, 15, 23, 28, 36

identify:

(i) the prime number,

..... [1]

(ii) the square number that is also a multiple of 4,

..... [1]

(iii) the irrational number.

..... [1]

17 Two numbers are given as $A = 2^2 \times 3 \times 5$ and $B = 2 \times 3^2 \times 7$.

(a) Find the Highest Common Factor (HCF) of A and B .

..... [2]

(b) Find the Lowest Common Multiple (LCM) of A and B . Give your answer as an integer.

..... [2]

18 (a) Express 72 as a product of its prime factors. Give your answer in index form.

..... [2]

(b) Write down the reciprocal of 72.

..... [1]

19 Consider the algebraic terms $X = 12a^2b$ and $Y = 30ab^3$, where a and b are distinct prime numbers.

(a) Find the Highest Common Factor (HCF) of X and Y in terms of a and b .

..... [2]

(b) Find the Lowest Common Multiple (LCM) of X and Y in terms of a and b .

..... [2]

20 (a) Write the number 30 040 008 in words using British standard formatting.

..... [1]

(b) Find the Highest Common Factor (HCF) of 45 and 105.

..... [2]

21 (a) Write the number forty-two billion seven hundred million and four in figures.

..... [1]

(b) From the following list of numbers:

$\sqrt{16}$, 27, $\frac{2}{3}$, π , 28, 45

identify:

(i) the cube number,

..... [1]

(ii) the triangle number,

..... [1]

(iii) the irrational number.

..... [1]

22 (a) Express 540 as a product of its prime factors. Give your answer in index form.

..... [2]

(b) Given that $540 \times k$ is a perfect square, where k is a positive integer, find the smallest possible value of k .

..... [2]

(c) Write down the reciprocal of k .

..... [1]

23 Two numbers, X and Y , are expressed as products of their prime factors:

$$X = 2^3 \times 3^2 \times 5 \times 11$$

$$Y = 2^2 \times 3^4 \times 7$$

(a) Find the Highest Common Factor (HCF) of X and Y . Give your answer as an integer.

..... [2]

(b) Find the Lowest Common Multiple (LCM) of X and Y . Give your answer in index form.

..... [2]

- 24** A security light flashes every 42 seconds. A second security light flashes every 60 seconds. Both lights flash together at 19:30.

(a) Find the Lowest Common Multiple (LCM) of 42 and 60.

..... [2]

(b) Determine the next time both lights will flash together.

..... [2]

-
- 25** (a) Write the number thirteen billion two hundred million and five in figures.

..... [1]

(b) A number N is expressed as the product of its prime factors as:

$$N = 2^x \times 3^y \times 5^z$$

The reciprocal of N has an indexing layout. Given that N is a perfect cube, a perfect square, and a multiple of 10, find the smallest possible integer value of N . Give your answer in index form.

..... [2]

(c) From the following set of numbers:

$$\{\sqrt[3]{27}, \sqrt{8}, \frac{22}{7}, 0.1313313331\dots, T_{10}\}$$

where T_{10} represents the 10th triangle number, identify:

(i) the value of the triangle number T_{10} ,

..... [1]

(ii) all the irrational numbers in the set.

..... [1]

26 Two algebraic terms, P and Q , are defined below, where x , y , and z are distinct prime numbers.

$$P = 126x^4y^2z^3$$

$$Q = 540x^2y^5$$

(a) Express 126 and 540 as products of their prime factors.

..... [2]

(b) Find the Highest Common Factor (HCF) of P and Q in terms of x , y , and z .

..... [2]

(c) Find the Lowest Common Multiple (LCM) of P and Q in terms of x , y , and z . Give your coefficient as an integer.

..... [2]

27 Three integers are given as $A = 2^3 \times 3^x \times 5^2$, $B = 2^y \times 3^2 \times 7$, and $C = 3^3 \times 5^z \times 11$.

- (a) Given that the Highest Common Factor of A and B is $\text{HCF}(A, B) = 36$, find the exact values of the exponents x and y .

..... [2]

- (b) Given that the Lowest Common Multiple of A and C is $\text{LCM}(A, C) = 2^3 \times 3^4 \times 5^3 \times 11$, find the exact values of the exponents x and z .

..... [2]

- (c) State whether the number $\frac{A \times B}{C}$ is rational or irrational when $x = 2, y = 2, z = 1$.

..... [1]

28 (a) Write the number 'five billion two hundred million and seventy-three' in figures.

..... [1]

- (b) Consider the following list of real numbers:

$\sqrt{5}$, 16, 27, 28, $\frac{7}{9}$, 41

Identify from the list:

- (i) the prime number,

..... [1]

(ii) the cube number,

..... [1]

(iii) the triangle number.

..... [1]

-
- 29** (a) Express the integer 360 as a product of its prime factors. Give your final answer in index form.

..... [2]

- (b) Write down the exact value of the reciprocal of 360.

..... [1]

-
- 30** Two numbers are given as $P = 2^3 \times 3 \times 5^2$ and $Q = 2 \times 3^2 \times 5$.

- (a) Find the Highest Common Factor (HCF) of P and Q . Give your answer as an integer.

..... [2]

- (b) Find the Lowest Common Multiple (LCM) of P and Q . Give your answer as an integer.

..... [2]

- 31 Consider the simple algebraic expressions $M = 14x^2y$ and $N = 35xy^3$, where x and y are distinct prime numbers.

- (a) Find the Highest Common Factor (HCF) of M and N in terms of x and y .

..... [2]

- (b) Find the Lowest Common Multiple (LCM) of M and N in terms of x and y .

..... [2]

- 32 (a) A specific natural number W is described in words as 'two billion four hundred and eighty-four million'. Express W as a product of its prime factors using index form.

..... [3]

- (b) A statistical tracking variable V updates continuous sample bins. The value of V is given by:

$$V = \frac{W \times 10^4}{2 \times 3^3 \times 5^6}$$

Evaluate the expression to show that V is a perfect square number.

[3]

-
- 33** Two complex variables inside a digital geometric rendering algorithm are defined by the prime factor layouts:

$$X = 2^a \times 3^4 \times 5^2 \times y^3$$

$$Y = 2^3 \times 3^b \times 5^c \times z^2$$

where y and z are distinct, unknown prime numbers greater than 5, and a , b , and c are integers.

- (a) Given that $\text{HCF}(X, Y) = 360$, calculate the exact values of a , b , and c .

..... [3]

- (b) Given that $\text{LCM}(X, Y) = 2.43 \times 10^7$, write down a mathematical equation relating the prime variables y and z .

..... [2]

-
- 34** A continuous automated monitoring station tracks data packet volumes transmitted across a global telecommunications network.

- (a) At peak capacity, one key transmission link records a data volume of thirty-four billion eight hundred million and ninety-two bytes per hour. Write this numerical quantity in figures.

..... [1]

- (b) A structural routing constant K is measured across the network data nodes and belongs to the following given set of numbers:

$$\{\sqrt{18}, 64, \frac{22}{7}, 45, 53\}$$

Identify from this set:

- (i) the prime number,

..... [1]

- (ii) the square number that is also a cube number,

..... [1]

- (iii) the irrational number.

..... [1]

- 35** A multi-stage mechanical sorting conveyor uses two large independent primary driving wheels. The structural indices for the tooth configurations are represented by the variables A and B , where:

$$A = 2^4 \times 3^2 \times 5 \times 11$$

$$B = 2^2 \times 3^3 \times 5^2 \times 7$$

- (a) Calculate the Highest Common Factor (HCF) of A and B . Give your answer as an exact

integer.

..... [2]

- (b) Calculate the Lowest Common Multiple (LCM) of A and B . Give your answer as a calculation evaluated to 3 significant figures, expressed in standard scientific notation.

..... [2]

-
- 36** An astronomical monitoring facility models periodic cycles of digital telemetry feeds. Satellite Alpha transmits data packets every 1.75 minutes, while Satellite Beta transmits packets every 2.25 minutes. Both systems log initial concurrent data pings at exactly 04:00.

- (a) Convert the time cycles into simple written fractions of minutes, and find the precise Lowest Common Multiple (LCM) of the periodic cycles for Satellite Alpha and Satellite Beta.

..... [2]

- (b) Calculate the total elapsed time in minutes, and determine the next exact time on a 24-hour clock when both satellites will transmit simultaneously.

..... [2]

-
- 37** (a) Evaluate the expression.

$$5^2 \times \sqrt[3]{8}$$

..... [2]

(b) Evaluate the expression.

$$10^3 - \sqrt{144}$$

..... [2]

38 (a) Evaluate the expression.

$$\sqrt{121} + 3^3 \times \sqrt[3]{1}$$

..... [2]

(b) Evaluate the expression.

$$\frac{2^3 \times \sqrt{169}}{10^2 - 96}$$

..... [2]

39 (a) Evaluate the expression.

$$\sqrt[3]{64} \times 4^2 - \sqrt{225}$$

..... [2]

(b) Evaluate the expression.

$$\frac{\sqrt{196}}{\sqrt[3]{8}} + 12^2$$

..... [2]

40 (a) Evaluate the expression.

$$2 \times \sqrt{81} - 3^2 \times \sqrt[3]{27}$$

..... [2]

(b) Evaluate the expression.

$$\sqrt{100} \times \sqrt{4} - 5^3$$

..... [2]

41 (a) Convert 0.35 into a proper fraction written in its simplest form.

..... [1]

- (b) Convert $\frac{7}{20}$ into a percentage.

..... [1]

- 42 (a) Convert 12.5% into a fraction written in its simplest form.

..... [1]

- (b) Write the mixed number $2\frac{3}{5}$ as a decimal number.

..... [1]

- 43 (a) Convert the improper fraction $\frac{14}{4}$ into a mixed number in its simplest form.

..... [1]

- (b) Convert your answer from part (a) into a decimal.

..... [1]

44 (a) Express 0.08 as a percentage.

..... [1]

(b) Express 0.08 as a proper fraction written in its simplest form.

..... [1]

45 (a) Convert 140% into an improper fraction written in its simplest form.

..... [1]

(b) Convert 140% into a decimal number.

..... [1]

46 (a) Convert 0.625 into a proper fraction written in its simplest form.

..... [1]

- (b) Express the mixed number $4\frac{11}{16}$ as a terminating decimal.

..... [2]

- 47 (a) Express 0.052% as a proper fraction written in its simplest form.

..... [2]

- (b) Evaluate the expression below, leaving your final answer as an improper fraction in its simplest form:

$$5.45 - \frac{3}{8} + 120\%$$

..... [2]

- 48 (a) An algebraic scale parameter is given by the expression:

$$K = \frac{0.36 \times 1.5}{0.08}$$

Evaluate K and express the result as an improper fraction in its simplest form.

..... [2]

- (b) Convert your fraction answer from part (a) into a percentage.

..... [1]

- 49 (a) Arrange the following real numbers in ascending order of magnitude. Show your supporting equivalence evaluations explicitly.

$$0.64, \quad \frac{5}{8}, \quad 62.8\%, \quad \frac{16}{25}$$

..... [3]

- (b) Write down the exact value of the reciprocal of $\frac{16}{25}$ as a mixed number in its simplest form.

..... [1]

- 50 (a) Show that the expression below simplifies to exactly 7.5×10^{-1} .

$$\frac{0.125 \times 12}{2}$$

[2]

- (b) Find a terminating decimal value that lies strictly between $\frac{7}{8}$ and 88%.

..... [2]

-
- 51 (a) An algebraic matrix scale parameter, M , is defined by the analytical expression:

$$M = \frac{0.0075 \times 3.2 \times 10^3}{4.8}$$

Evaluate M and express its structural value as an improper fraction in its simplest form.

..... [2]

- (b) Express your fraction answer from part (a) as a percentage.

..... [1]

- (c) Show that the reciprocal of M can be written exactly as a terminating decimal in the form $a \times 10^{-1}$, where a is an integer.

[2]

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- 52 A continuous experimental tracking array archives variable data bins across four specific channels. The current data densities are given by:

$$A = 0.8125, \quad B = \frac{13}{16}, \quad C = 81.3\%, \quad D = \frac{41}{50}$$

- (a) Show analytically that channel A and channel B possess exact structural equivalence.

[2]

- (b) Arrange all four data density values in ascending order of magnitude. Show your supporting equivalence evaluations explicitly.

..... [3]

- 53 A structural processing index P within a computer design interface is governed by the expression:

$$P = \frac{2.75 - \frac{3}{20}}{1.3 \times 4}$$

- (a) Calculate the exact value of P . Give your answer as a proper fraction written in its simplest form.

..... [3]

- (b) Convert your fraction from part (a) into a percentage.

..... [1]

- (c) Find a terminating decimal value that lies strictly between the value of P and 50.1%.

..... [2]

- 54** (a) Arrange the following list of rational numbers in ascending order of magnitude.

$$0.42, \quad \frac{2}{5}, \quad 45\%, \quad \frac{9}{20}$$

..... [2]

- (b) Insert the correct relational symbol ($>$ or $<$ or $=$) into the statement below to make it true.

$$-\frac{3}{4} \quad \text{-----} \quad -0.8$$

..... [1]

- 55** (a) Arrange the following real numbers in descending order of magnitude.

$$\frac{3}{8}, \quad 0.35, \quad 32\%, \quad \frac{7}{20}$$

..... [2]

- (b) Write down the structural inequality symbol ($\geq$ or $\leq$) that correctly satisfies the given relationship.

$$\frac{12}{3} \quad \text{-----} \quad 2^2$$

..... [1]

- 56 (a)** Arrange the following quantities in ascending order of magnitude.

$$1.25, \quad 122\%, \quad \frac{5}{4}, \quad \frac{6}{5}$$

..... [2]

- (b)** State whether the following mathematical statement is true or false.

$$\frac{1}{8} \neq 0.125$$

..... [1]

- 57 (a)** Arrange the following numeric elements in descending order of magnitude.

$$0.08, \quad \frac{7}{100}, \quad 6.5\%, \quad \frac{3}{40}$$

..... [2]

- (b)** Identify which of the following expressions has the smallest magnitude.

$$2^{-1}, \quad 0.4, \quad \frac{1}{3}$$

..... [1]

-
- 58 (a)** Identify all the numbers in the list below that are strictly greater than 0.55.

54.5%, $\frac{11}{20}$, $\frac{3}{5}$, 0.505

..... [2]

- (b)** Complete the analytical statement using one of the given structural symbols: =, $\neq$, >, <.

$\frac{14}{25}$ ----- 56%

..... [1]

-
- 59 (a)** Arrange the following quantities in ascending order of magnitude.

0.66, $\frac{5}{8}$, 62.8%, $\frac{13}{20}$

..... [2]

- (b)** Insert the correct structural inequality or relational symbol (>, <, or =) to make the statement true.

4^{-2} ----- 0.0625

..... [1]

- 60** (a) Arrange the following algebraic index values in descending order of magnitude.

$$\left(\frac{4}{9}\right)^{\frac{1}{2}}, \quad 0.65, \quad 66\%, \quad \frac{5}{8}$$

..... [2]

- (b) Write down the correct relational symbol ($>$, $<$, or $=$) that satisfies the analytical expression.

$$-1.4 \quad \text{-----} \quad -\frac{7}{5}$$

..... [1]

- 61** (a) Arrange the following real terms in ascending order of magnitude.

$$\frac{17}{25}, \quad 67.5\%, \quad 0.682, \quad \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$$

..... [2]

- (b) Complete the expression using the correct symbol from the set $\{>, <, \geq, \leq, =\}$.

$$3.5\% \quad \text{-----} \quad \frac{7}{200}$$

..... [1]

-
- 62 (a)** Identify which expression in the following list has the absolute largest magnitude.

$$-0.75, \quad -\frac{4}{5}, \quad -82\%, \quad -\frac{7}{8}$$

..... [2]

- (b)** Insert the single relational symbol ($>$ or $<$ or $=$) that makes the structural comparison true.

$$\left(\frac{27}{8}\right)^{-\frac{1}{3}} \quad \text{-----} \quad 0.6$$

..... [1]

-
- 63 (a)** Identify which expression in the following list has the absolute largest magnitude.

$$-0.75, \quad -\frac{4}{5}, \quad -82\%, \quad -\frac{7}{8}$$

..... [2]

- (b)** Insert the single relational symbol ($>$ or $<$ or $=$) that makes the structural comparison true.

$$\left(\frac{27}{8}\right)^{-\frac{1}{3}} \quad \text{-----} \quad 0.6$$

..... [1]

64 The following four quantities are written in different numerical formats:

$$A = \frac{17}{20}, \quad B = 84.5\%, \quad C = 0.844, \quad D = \sqrt{\frac{18}{25}}$$

Use your graphics calculator to convert these values.

(a) Write down the value of D correct to 4 decimal places.

..... [1]

(b) Arrange the four quantities A , B , C , and D in order of magnitude, starting with the smallest.

..... [2]

(c) Complete the statement using one of the relational symbols $=$, $\neq$, $>$, or $<$ to make it correct.

$$A \text{ ————— } \frac{85}{100}$$

..... [1]

65 Consider the four expressions shown below:

$$P = \pi - 2.3, \quad Q = \frac{21}{25}, \quad R = \sqrt{0.71}, \quad S = 84.1\%$$

(a) Find the exact decimal value of Q .

..... [1]

(b) Write down the value of P correct to 3 significant figures.

..... [1]

(c) Arrange the four expressions P , Q , R , and S in ascending order of magnitude.

..... [2]

66 Five numbers are given by the expressions below:

$$V_1 = \frac{7}{8}, \quad V_2 = 0.87, \quad V_3 = \sqrt{0.76}, \quad V_4 = 87.2\%, \quad V_5 = \frac{\pi}{3.6}$$

(a) Find the value of V_5 correct to 3 significant figures.

..... [1]

(b) Order the five expressions by magnitude, from smallest to largest.

..... [2]

(c) Insert the correct relational symbol ($\leq$, $\geq$, $<$, or $>$) to make the inequality true:

$$V_1 \text{ ______ } V_4$$

..... [1]

67 Five numbers are positioned along an interval of a real number line:

$$K = 0.63, \quad L = \frac{14}{22}, \quad M = 63.5\%, \quad N = \sqrt{0.4}, \quad P = \frac{\pi}{5}$$

(a) Calculate the value of P correct to 4 decimal places.

..... [1]

(b) Order the five numbers from smallest to largest magnitude.

..... [2]

(c) Complete the statement below using the most appropriate relational symbol from the set $\{<, =, >\}$.

$$L \text{ ______ } M$$

..... [1]

-
- 68** Four locations on a coastal map have heights above sea level, in metres, determined by multi-stage geometric observations:

$$P = 3 \sin(24^\circ) + \tan(12^\circ), \quad Q = \frac{71}{50}, \quad R = \sqrt{2.01} - 0.003, \quad S = 141.6\%$$

- (a) Calculate the value of P correct to 4 significant figures.

..... [1]

- (b) Arrange the four locations P , Q , R , and S in ascending order of their magnitudes.

..... [2]

- (c) Complete the statement using either $<$ or $>$ to make it true:

$$P \text{ ______ } R$$

..... [1]

-
- 69** Consider the following four rational and exponential expressions:

$$P = \log_3(27), \quad Q = \left(\frac{1}{16}\right)^{-\frac{3}{4}}, \quad R = \left(\frac{25}{4}\right)^0, \quad S = \sqrt{\frac{81}{4}}$$

- (a) Find the exact value of Q as an integer.

..... [1]

- (b) Arrange the four numbers P , Q , R , and S in order of magnitude, starting with the largest.

..... [2]

- (c) Fill in the gap with the correct relational symbol ($<$ or $>$) to validate the expression:

$$P \text{ ______ } S$$

..... [1]

- 70** Five quantities are determined analytically by the algebraic properties shown below:

$$X_1 = \log_5\left(\frac{1}{25}\right), \quad X_2 = -\sqrt{\frac{9}{4}}, \quad X_3 = \tan(225^\circ), \quad X_4 = \left(\frac{8}{27}\right)^{-\frac{2}{3}}, \quad X_5 = 125\%$$

- (a) Show that the expression for X_4 evaluates precisely to $\frac{9}{4}$.

[2]

(b) Arrange all five quantities in ascending order of magnitude.

..... [2]

(c) Insert the correct relational symbol ($<$ or $>$) to complete the inequality:

$$X_2 \text{ ______ } X_1$$

..... [1]

71 Work out the value of the following expressions.

(a) $12 - 4 \times 5 + 3$

..... [1]

(b) $24 \div (2 \times 3) + 7$

..... [1]

72 Work out the value of the following calculations, giving your answers as fractions in their simplest form.

(a) $\frac{3}{5} + \frac{1}{10}$

..... [1]

(b) $\frac{4}{7} \times \frac{14}{15}$

..... [1]

73 Work out the value of the following mixed number arithmetic operations, giving your answers as mixed numbers in their simplest form.

(a) $2\frac{1}{3} - 1\frac{5}{6}$

..... [2]

(b) $1\frac{1}{4} \div \frac{3}{8}$

..... [2]

74 Work out the value of the following decimal operations without using a calculator.

(a) $3.45 + 1.8$

..... [1]

(b) 0.6×0.04

..... [1]

75 Work out the value of the following expressions, giving each answer as an exact integer or fraction in its simplest form.

(a) $5 - 3 \times (-2)^3 + 18 \div (-3)$

..... [2]

(b) Evaluate the expression below.

$$\frac{4 \times (7 - 12) + (-3)^2}{22 - 3 \times 5}$$

..... [2]

76 Evaluate the following expressions involving mixed numbers, fractions, and integers, giving your final answers as mixed numbers in their lowest terms.

(a) $3\frac{1}{4} - 1\frac{2}{3} \times \frac{6}{7}$

..... [2]

(b) Show that the expression below evaluates to $2\frac{2}{5}$.

$$\left(2\frac{1}{2} + \frac{7}{10}\right) \div 1\frac{1}{3}$$

[2]

77 (a) Evaluate.

$$5 \times (3 - 8) + 14$$

..... [2]

(b) Work out.

$$4.5 + 1.2 \div 0.3$$

..... [2]

78 (a) Write down the value of.

$$(-4) \times (-2) \times (-3)$$

..... [1]

- (b) Work out, giving your answer as a fraction in its simplest form.

$$1\frac{3}{4} + \frac{5}{8}$$

..... [2]

- 79 (a) Work out.

$$\frac{3.5 - 1.1}{0.4}$$

..... [2]

- (b) Find the exact value of.

$$\frac{5}{12} \times \frac{4}{15}$$

..... [2]

- 80 (a) Insert brackets into the statement below to make it correct.

$$3 + 5 \times 2 - 4 = 12$$

..... [1]

(b) Work out.

$$\left(-\frac{2}{5}\right) \times \left(-\frac{3}{4}\right)$$

..... [2]

81 (a) Calculate.

$$7 - 3 \times 4 + 2$$

..... [2]

(b) Find the value of.

$$0.05 \times 400$$

..... [1]

82 Simplify each of the following expressions, leaving your answers in index form.

(a) $3^4 \times 3^2$

..... [1]

(b) $\frac{5^7}{5^4}$

..... [1]

(c) $(2^3)^4$

..... [1]

83 Evaluate each of the following numerical expressions. Give your answers as integers or fractions in their simplest form.

(a) 7^0

..... [1]

(b) 4^{-2}

..... [1]

(c) 3×2^{-3}

..... [1]

84 Find the exact value of each expression.

(a) $25^{\frac{1}{2}}$

..... [1]

(b) $27^{\frac{1}{3}}$

..... [1]

(c) $16^{\frac{3}{4}}$

..... [2]

85 Simplify the following operations completely.

(a) $2^{-4} \times 2^6$

..... [1]

(b) $\frac{6^2}{6^{-1}}$

..... [1]

- (c) Show that the expression $\left(\frac{2}{3}\right)^{-2}$ reduces exactly to $2\frac{1}{4}$.

[2]

- 86** Work out the value of each expression, giving your answer as a single integer or a simplified fraction.

(a) $8^{-\frac{1}{3}}$

..... [2]

(b) $\left(\frac{9}{4}\right)^{-\frac{1}{2}}$

..... [2]

- 87** Evaluate each of the following expressions completely, giving your final answer as an exact integer or a fraction in its simplest form.

(a) $\left(\frac{1}{8}\right)^{-\frac{2}{3}}$

..... [2]

(b) $125^{\frac{1}{3}} \times 2^{-3}$

..... [2]

88 Solve the following index equations for x .

(a) $3^x = \frac{1}{81}$

..... [2]

(b) $4^{2x-1} = 32$

..... [2]

89 Simplify the operational expressions.

(a) Find the exact value of the expression.

$$\frac{2^5 \times 6^3}{3^2 \times 4^3}$$

..... [3]

(b) Show that the expression reduces exactly to an integer.

$$\frac{\left(5^{\frac{1}{2}}\right)^4 \times 10^2}{2^2 \times 5^3}$$

[2]

90 Work out the value of the following composite expressions.

(a) $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$

..... [2]

(b) $\left(4^{\frac{3}{2}} + 27^{\frac{2}{3}}\right)^{-1}$

..... [3]

91 Evaluate each of the operations without using a calculator, leaving answers as standard rational values.

(a) $\left(2\frac{7}{9}\right)^{-\frac{1}{2}}$

..... [2]

(b) $\frac{3^{-5} \times 9^2}{27^{-1}}$

..... [2]

92 Solve the simultaneous index equations for x and y .

$$2^x \times 4^y = \frac{1}{8}$$

$$27^x \times 3^y = 9^{\frac{1}{2}}$$

..... [4]

93 An advanced engineering formula determines the operational efficiency metric, E , based on an input index variable x such that:

$$E = \frac{243^{\frac{x}{5}} \times 4^{-x}}{3^{x-1}}$$

(a) Show analytically that the expression for E simplifies exactly to 3×4^{-x} .

[2]

(b) Find the value of E when $x = 2.7$, correct to 3 significant figures.

..... [2]

94 Convert each of the following standard form values into ordinary numbers.

(a) 3.08×10^4

..... [1]

(b) 6.1×10^{-3}

..... [1]

95 Calculate the value of the operational expressions, giving your answers in standard form.

(a) $(2 \times 10^3) \times (4 \times 10^5)$

..... [2]

(b) $\frac{9 \times 10^2}{3 \times 10^6}$

..... [2]

-
- 96** Evaluate the expression below, converting the intermediate step out of ordinary units to give your final answer exactly in standard form.

$$(5 \times 10^4) \times (3 \times 10^{-2})$$

..... [2]

-
- 97** Work out the value of each calculation, giving your answers in standard form.

(a) $(6 \times 10^7) \times (5 \times 10^{-3})$

..... [2]

(b) $\frac{2 \times 10^{-4}}{8 \times 10^{-2}}$

..... [2]

-
- 98** Evaluate the following expression, giving your final answer in standard form.

$$4.3 \times 10^5 - 7 \times 10^4$$

..... [2]

99 Find the value of k in each of the following statements.

(a) $0.000045 \times 200 = 9 \times 10^k$

..... [2]

(b) $\frac{1}{250000} = 4 \times 10^k$

..... [2]

100 Solve the following question regarding operations in standard form.

(a) Show that the sum of 2.4×10^{-3} and 5.8×10^{-4} can be written exactly as 2.98×10^{-3} .

[2]

(b) Calculate the value of $(1.2 \times 10^4)^2$. Give your answer in standard form.

..... [2]

101 Calculate the exact value of the expression, giving your final answer in standard form notation.

$$\frac{(3.5 \times 10^{-5}) \times (4 \times 10^9)}{2 \times 10^2}$$

..... [3]

102 Given that $P = 4 \times 10^n$ and $Q = 5 \times 10^{n-1}$, where n is an integer.

(a) Show that the sum $P + Q$ can be written exactly in standard form as 4.5×10^n .

[2]

(b) Express the quotient $\frac{P^2}{Q}$ in standard form, leaving the index of the base 10 in terms of n .

..... [2]

103 Evaluate the following expression without using a calculator. Give your final answer in standard form.

$$\frac{(1.2 \times 10^{-3}) + (8 \times 10^{-4})}{4 \times 10^{-8}}$$

..... [3]

104 Solve the following question regarding standard form transformations.

(a) Solve for the integer k such that:

$$(2.5 \times 10^k)^2 \times (8 \times 10^3) = 5 \times 10^2$$

..... [3]

(b) Write the value of $(5 \times 10^4)^{-3}$ exactly in standard form.

..... [2]

105 Consider the standard form variables $X = 3 \times 10^{-5}$ and $Y = 9 \times 10^{-6}$.

(a) Calculate $X - Y$, giving your answer in standard form.

..... [2]

(b) Evaluate the cube root of the quotient expression shown below, leaving your answer in standard form.

$$\sqrt[3]{\frac{X^2}{Y}}$$

..... [3]

-
- 106** An exact analytical evaluation requires simplifying a multi-stage fraction into standard form without a calculator.

- (a) Show that the following expression simplifies exactly to 2.5×10^{-1} .

$$\frac{(5 \times 10^{-3})^2}{2 \times 10^{-4} - 1 \times 10^{-4}}$$

[2]

- (b) Find the exact standard form value of M if:

$$\frac{1}{M} = (4 \times 10^3) + (1 \times 10^2)$$

..... [3]

-
- 107** An astronomical observatory records continuous statistics tracking regarding cosmic radiation. The total energy captured by a sensor during an initial observation phase is 4.2×10^7 joules. During a second phase, the sensor records an additional 1.9×10^6 joules.

- (a) Calculate the total combined energy recorded across both phases. Give your answer in standard form.

..... [2]

- (b) The total combined energy is distributed evenly across 5.0×10^3 storage blocks. Calculate the energy allocated to each block, giving your answer in standard form.

..... [2]

-
- 108** A geological software tracking model estimates the mass of a large rock formation to be 5.82×10^8 kilograms. A minor seismic event separates a section of rock with a mass of 9.4×10^6 kilograms.

- (a) Calculate the remaining mass of the main rock formation. Give your answer in standard form, correct to 3 significant figures.

..... [2]

- (b) Express the separated mass as a percentage of the original total mass.

..... [2]

-
- 109** In a laboratory test, a computer simulation records the total number of data operations executed by a parallel processing network. System A performs 7.5×10^{11} operations and System B performs 2.5×10^{10} operations.

- (a) Calculate how many times greater the number of operations performed by System A is compared to System B.

..... [2]

- (b) Find the average number of operations performed by these two systems, leaving your answer in standard form.

..... [2]

-
- 110** Continuous statistics tracking of an online cloud-computing database monitors the total volume of incoming data traffic. Over one financial quarter, Database Alpha processes 4.76×10^{14} bytes of data, while Database Beta processes 8.95×10^{12} bytes of data.

- (a) Calculate the total combined data volume processed by both databases during this quarter. Give your answer in standard form, correct to 3 significant figures.

..... [2]

- (b) Determine the number of times greater the data volume processed by Database Alpha is compared to Database Beta. Give your answer correct to 3 significant figures.

..... [2]

-
- 111** An industrial factory uses a multi-stage formula to model the mass extraction of refined silicon from quartz ore. The total mass of ore processed annually is 2.48×10^7 tonnes. The mass of refined silicon yielded, M tonnes, is given by the formula:

$$M = \frac{(2.48 \times 10^7) \times (3.14 \times 10^{-3})}{1.25 \times 10^2}$$

(a) Calculate the value of M . Give your answer in standard form, correct to 3 significant figures.

..... [3]

(b) Convert your answer from part (a) into an ordinary number.

..... [1]

112 Write the number 483.7062 correct to the specified degrees of accuracy.

(a) Correct to 2 decimal places.

..... [1]

(b) Correct to 3 significant figures.

..... [1]

113 Estimate the value of the following calculation by rounding each initial number correct to exactly 1 significant figure. Show your intermediate rounding step clearly.

$$\frac{39.4 \times 4.87}{0.193}$$

..... [3]

114 Round each given value according to the specified limits of accuracy.

(a) Write 0.003 049 correct to 2 significant figures.

..... [1]

(b) Write 79 512 correct to 1 significant figure.

..... [1]

115 Given the mathematical expression:

$$\frac{41.3 \times 0.0487}{\sqrt{0.0384}}$$

(a) Estimate the value of the expression by rounding each initial number correct to exactly 1 significant figure. Show your working clearly.

..... [2]

(b) Write down the number 0.0487 correct to 2 decimal places.

..... [1]

116 An analytical model evaluates the volumetric structural index of an alloy component.

(a) Show that making a structural estimate of the calculation below by rounding each value to 1 significant figure reduces the expression to exactly 0.06.

$$\frac{5.84 \times 0.203}{187.9}$$

[2]

(b) Express your answer to part (a) as a fraction in its simplest terms.

..... [1]

117 A total financial sum of £140 is shared among three siblings in the ratio 2 : 3 : 5.

(a) Calculate the share received by the sibling who gets the largest amount.

..... [2]

- (b) Calculate the difference in value between the largest and the smallest shares.

..... [2]

- 118** A specific baking recipe requires 120 g of flour to create a batch of 8 identical scones.

- (a) Calculate the mass of flour required to bake exactly 20 of these scones.

..... [2]

- (b) Determine the maximum number of full complete scones that can be baked using 450 g of flour.

..... [2]

- 119** A modern alloy compound is composed by mixing copper, zinc, and nickel in the mass ratio 7 : 4 : 1.

- (a) Given that a particular block of alloy contains 14 kg of copper, calculate the mass of zinc present.

..... [2]

- (b) Calculate the total mass of an alloy block that contains exactly 3.5 kg of nickel.

..... [2]

120 A large container contains a total fluid volume of 1800 ml of fruit punch, composed of orange juice, pineapple juice, and lime juice. The quantities are mixed in the ratio 5 : 3 : 1.

(a) Show that the volume of pineapple juice used is exactly 600 ml.

[2]

(b) Due to a recipe alteration, the volume of lime juice is doubled while the other two components remain constant. Write down the new simplified integer ratio of orange juice to pineapple juice to lime juice.

..... [2]

121 An industrial mixing tank combines chemicals X and Y such that the mass of chemical X to the mass of chemical Y is in the ratio 3 : 7.

(a) Express the mass of chemical X as a standard fraction of the total mass of the mixture.

..... [1]

(b) If the mass of chemical Y in the tank exceeds the mass of chemical X by exactly 24 kg, calculate the total combined mass of the mixture inside the tank.

..... [3]

122 The scale model of a historical cruise ship is built using an exact linear scale of 1 : 150.

- (a) The physical mast of the actual cruise ship is 30 m high. Calculate the height of the mast on the scale model in centimetres.

..... [2]

- (b) The total deck surface area of the scale model is 400 cm^2 . Find the actual deck surface area of the real cruise ship, giving your final answer in square metres.

..... [3]

123 Write the ratio 45 cm : 1.2 m in its simplest constituent integers.

..... [2]

124 A chemistry laboratory technician needs to prepare a specific saline solution. A stock mixture of volume V ml contains water and salt in the volume ratio 7 : 2.

- (a) Find the fraction of the stock mixture that is pure water.

..... [1]

- (b) The technician adds 60 ml of pure water to the stock mixture. The new modified volume ratio of water to salt becomes 4 : 1. Show that the initial volume V of the stock mixture satisfies the equation:

$$\frac{7}{9}V + 60 = \frac{8}{9}V$$

[2]

- (c) Calculate the initial volume V of the stock mixture.

..... [2]

- 125** An architectural draughtsman produces a layout drawing of a commercial office floor using a standard linear scale of 1 : n .

- (a) A rectangular conference hall with real physical dimensions of 12 m by 16 m is represented on the blueprint by an area of 48 cm². Show that the value of n is exactly 200.

[3]

- (b) Using this exact blueprint scale, calculate the map length, in centimetres, of a corridor that has an actual physical length of 54 m.

..... [2]

126 A business partnership allocates annual bonus payouts to three senior managers, *A*, *B*, and *C*, in a shifting ratio based on production points. Initially, £3600 is split in the ratio 3 : 4 : 5.

(a) Calculate the financial amount allocated to manager *B* under this initial distribution.

..... [2]

(b) Due to an evaluation review, the distribution ratio is revised to 2 : 3 : 4 for the same total sum of £3600. Determine which manager loses out the most financially, and calculate the exact magnitude of their financial loss.

..... [3]

127 (a) An aeroplane travels a distance of 1400 km at an average speed of 400 km/h. Calculate the time taken for this journey, giving your answer in hours and minutes.

..... [2]

(b) A second aeroplane travels the same distance of 1400 km. The journey takes 2 hours and 20 minutes.
Calculate the average speed of this aeroplane, giving your answer in km/h.

..... [2]

- 128** (a) A cyclist travels at an average speed of 6 m/s for exactly 15 minutes. Calculate the distance travelled by the cyclist, giving your answer in kilometres.

..... [3]

- (b) Show that an average speed of 6 m/s is exactly equivalent to 21.6 km/h.

[2]

-
- 129** (a) A motorist drives a distance of 45 km. The journey takes 3 hours and 45 minutes. Calculate the average speed of the motorist in km/h.

..... [2]

- (b) The motorist's car consumes fuel at a rate of 6 litres per 100 km driven. Calculate the amount of fuel used during this 45 km journey.

..... [2]

-
- 130** (a) A cyclist travels the first 12 km of a journey at an average speed of 16 km/h and the remaining 24 km at an average speed of 20 km/h. Calculate the average speed of the cyclist for the entire

journey.

..... [3]

- (b) A container has a volume of 4000 cm^3 . It is filled with a liquid that has a uniform density of 1.15 g/cm^3 . Calculate the total mass of the liquid in kilograms.

..... [2]

-
- 131** (a) An express train covers a total distance of d km. It travels the first 90 km at an average speed of 120 km/h. The train completes the rest of the journey in 1 hour and 45 minutes at an average speed of 100 km/h.
Calculate the total distance d of the journey.

..... [2]

- (b) Calculate the average speed of the express train for the entire journey.

..... [2]

-
- 132** (a) A motorist drives from town A to town B at an average speed of 60 km/h, and then returns along the same route from town B to town A at an average speed of 90 km/h. Show that the average speed for the entire two-way journey is exactly 72 km/h.

[3]

- (b) The total driving time for the entire two-way journey is 3 hours and 20 minutes. Calculate the distance between town A and town B.

..... [2]

-
- 133** (a) A business consultant works in London and earns £45 per hour. One week, the consultant works for 36 hours. Calculate the consultant's total earnings for that week.

..... [1]

- (b) The consultant exchanges $\frac{2}{5}$ of these total weekly earnings into Euros (€) at a financial institution where the exchange rate is £1 = €1.16. Calculate the amount of Euros received.

..... [2]

- (c) A car used by the consultant consumes fuel at a rate of 8 litres per 100 km. Fuel costs €1.45 per litre. Calculate the cost of the fuel required to drive a distance of 250 km.

..... [2]

-
- 134** (a) A courier travels from town A to town B at an average speed of v km/h, and immediately returns along the same route from town B to town A at an average speed that is 15 km/h slower. Show that the average speed for the entire two-way journey is given by the expression:

$$\frac{2v^2 - 30v}{2v - 15}$$

[3]

- (b) Given that the average speed for the entire two-way journey is 40 km/h, calculate the value of v .

..... [3]

-
- 135** (a) A commercial transport vehicle travels the first $\frac{1}{3}$ of its total journey distance at a constant average speed of x km/h, and the remaining $\frac{2}{3}$ of the distance at a constant average speed of $2x$ km/h. Show that the average speed for the entire journey is exactly $\frac{3}{2}x$ km/h.

[3]

- (b) The total distance of the journey is 270 km. The total time taken for the journey is 3 hours and 45 minutes. Calculate the value of x .

..... [2]

-
- 136** Alistair drives a total distance of 284 km. He drives the first 114 km at an average speed of 76 km/h. He completes the remaining journey in 2 hours 18 minutes.

(a) Calculate the time taken, in hours and minutes, for the first part of his journey.

..... [2]

(b) Calculate the average speed for the entire 284 km journey. Give your answer in km/h correct to 3 significant figures.

..... [3]

-
- 137** Fiona travels from London to Paris. She changes GBP 450 into EUR at an exchange rate of 1 GBP = 1.162 EUR.

(a) Calculate the amount of EUR Fiona receives.

..... [2]

- (b) In Paris, she buys a ticket for a high-speed train. The train travels a distance of 305 km and takes 1 hour 22 minutes. Calculate the average speed of the train in m/s.

..... [3]

-
- 138** An extraction plant processes raw materials across international facilities. The operational cost functions rely on exchange dynamics between GBP, USD, and CAD. The current market rates indicate:

$$1 \text{ GBP} = 1.284 \text{ USD}$$

$$1 \text{ CAD} = 0.732 \text{ USD}$$

- (a) Convert GBP 3850 directly into CAD. Give your answer to the nearest integer currency unit.

..... [3]

- (b) A supply vehicle at the Canadian plant consumes fuel at a rate of 14.8 litres per 100 km. Fuel costs CAD 1.64 per litre. Calculate the total cost, in GBP, for a delivery run covering a distance of 645 km. Give your answer correct to 2 decimal places.

..... [4]

-
- 139** An antique vase is purchased for 250 GBP.

- (a) The value of the vase increases by 12% after one year. Calculate its new value.

..... [2]

- (b) Due to a minor scratch, the new value then decreases by 10% . Calculate the final value of the vase.

..... [2]

- 140** A rare collector card is sold for 180 EUR, which represents a profit of 20% on the original purchase price. Calculate the original purchase price.

..... [3]

- 141** The mass of a breeding colony of insects increases by 10% each week. At the start of the first week, the mass is 500 grams.

- (a) Calculate the mass of the colony at the end of 2 weeks.

..... [3]

- (b) Express the mass at the end of 2 weeks as a percentage of the original mass.

..... [1]

142 An investor buys shares in a company for 800 GBP.

- (a) In the first year, the value of the shares increases by 25% . Calculate the value of the shares at the end of the first year.

..... [2]

- (b) In the second year, the value of the shares decreases by 15% from their value at the end of the first year. Calculate the overall percentage increase from the original purchase price to the end of the second year.

..... [3]

143 Liam deposits 2400 EUR into a special investment account. The account accrues compound interest at a fixed rate of 5% per annum.

- (a) Without relying on a pre-programmed formula, show that the total value of the investment at the end of 2 years is 2646 EUR.

[2]

- (b) At the end of the third year, the total value of the investment is 2778.30 EUR. Calculate the amount of interest earned during the third year only.

..... [2]

144 A collector purchases a rare painting as an investment. Over the first year, its value increases by $a\%$. Over the second year, its value decreases by $a\%$ of its new value. The final value of the painting at the end of the second year is exactly $\frac{24}{25}$ of the original purchase price.

(a) Show that the value of a satisfies the equation:

$$\left(\frac{a}{100}\right)^2 = \frac{1}{25}$$

[3]

(b) Hence, find the value of a .

..... [1]

(c) The final value of the painting at the end of the second year is 48000 GBP. Calculate the original purchase price.

..... [2]

145 An investor opens a savings account with an initial deposit of 4500 GBP. The account accrues simple interest at a fixed annual rate of 3.85% over a period of 6 years.

(a) Calculate the total amount of simple interest earned by the investor at the end of the 6-year period.

..... [2]

- (b) Find the final value of the investment when the 6 years expire.

..... [1]

- 146** A high-yield corporate fund was valued at 125000 USD at the beginning of the financial fiscal term. Over a continuous timeframe of 3 years, the asset value grows at a compound interest rate of 4.25% per annum.

- (a) Calculate the cumulative value of the high-yield corporate fund at the termination of the 3-year term. Give your answer correct to 3 significant figures.

..... [3]

- (b) Calculate the total compound interest earned by the fund during this 3-year term. Give your answer correct to 3 significant figures.

..... [1]

- 147** An investor opens a fixed-yield corporate portfolio with an initial deposit of 14500 EUR. The capital appreciates at a compound interest rate of 4.35% per year.

- (a) Calculate the total value of the portfolio after 6 years, correct to the nearest cent.

..... [2]

- (b) At the end of the 6th year, a one-off structural asset management fee of 3.8% is deducted from the entire accumulated portfolio value. Calculate the net final value of the investor's portfolio.

..... [2]

- (c) Express the net final value found in part (b) as a percentage of the initial 14500 EUR deposit.

..... [2]

- 148** A construction conglomerate allocates capital into two separate financial engineering instruments, Account X and Account Y, splitting a total initial funding of 850000 AUD.

- (a) Account X yields simple interest at a fixed nominal rate of 5.65% per annum. Calculate the total interest earned if 480000 AUD is held in Account X for exactly 7 years and 3 months.

..... [3]

- (b) The remaining 370000 AUD is allocated to Account Y, which accumulates interest compounding continuously/annually at a rate of 4.95% per year. Show that the total value accumulated in Account Y after n years is given by the expression:

$$370000 \times (1.0495)^n$$

[1]

- (c) Find the minimum integer value of n required for the value of Account Y to exceed 600000

AUD.

..... [2]

149 A flight departs from Paris at 22 45 local time on Tuesday. The total flight duration is 7 hours and 30 minutes.

(a) Calculate the local arrival time in Paris using the 24-hour clock.

..... [2]

(b) State the day of the week on which the flight arrives in Paris.

..... [1]

150 The local time in Dubai is 4 hours ahead of the local time in London. A business executive in London sets up a video conference to begin at 14 15 local London time.

(a) Determine the local time in Dubai when the video conference begins using the 24-hour clock.

..... [1]

- (b) The video conference finishes at 16 05 local London time. Calculate the total duration of the conference in minutes.

..... [2]

- 151** An international cargo ship leaves Port A at 15 20 local time on 10 April. It arrives at Port B after a continuous voyage lasting exactly 3 days and 5 hours. The local time at Port B is 2 hours behind the local time at Port A.

- (a) Calculate the local time at Port B when the ship departs from Port A.

..... [1]

- (b) Calculate the local time and date at Port B when the ship arrives.

..... [2]

- 152** An aircraft leaves London (GMT) at 22 45 on a Tuesday.
The aircraft flies to Tokyo, which is in a time zone 9 hours ahead of GMT.
The total flight duration is 11 hours and 35 minutes.

- (a) Write the departure time of the aircraft using the 12-hour clock format, including p.m. or a.m.

..... [1]

(b) Calculate the local time and day in Tokyo when the aircraft lands.

(i) Local time

..... [2]

(ii) Day

..... [1]

153 A flight departs from London Heathrow Airport (Time Zone: UTC +0) on a Monday at 21 45 local time. The flight travels to Tokyo Haneda Airport (Time Zone: UTC +9). The total flight duration is 13 hours and 35 minutes.

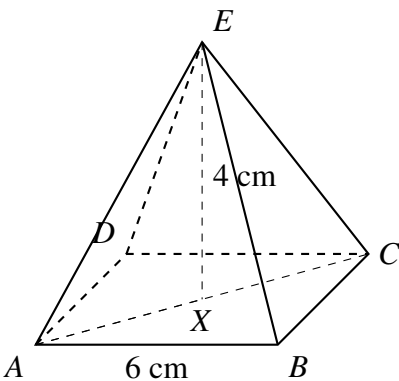
(a) Find the day and local time in Tokyo when the flight arrives. Give your answer using the 24-hour clock format.

..... [2]

(b) An analyst records the flight duration in hours. Express 13 hours and 35 minutes as a decimal number of hours, rounded to 3 significant figures.

..... [2]

- 154** A right square-based pyramid has vertices A, B, C, D on the base. The vertex E is directly above the centre of the square base, X . The side of the square base is 6 cm and the vertical height EX is 4 cm.



- (a) Calculate the exact length of AC .

..... [2]

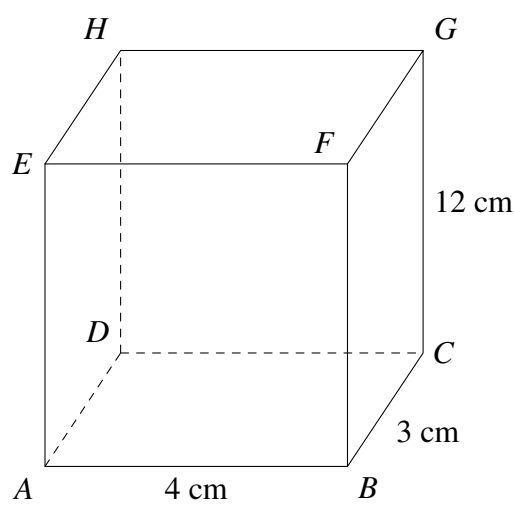
- (b) Calculate the exact length of AX .

..... [1]

- (c) Show that the length of the edge AE is exactly $\sqrt{34}$ cm.

[2]

- 155** The diagram shows a cuboid with rectangular base $ABCD$ and top $EFGH$.
 AE , BF , CG and DH are vertical edges.
 $AB = 4$ cm, $BC = 3$ cm and $CG = 12$ cm.



- (a) Calculate the length AC .

..... [2]

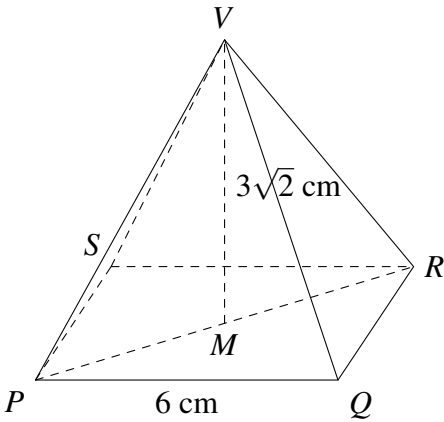
- (b) Calculate the length AG .

..... [2]

- (c) Show that the sine of the angle between the line AG and the plane $ABCD$ is $\frac{12}{13}$

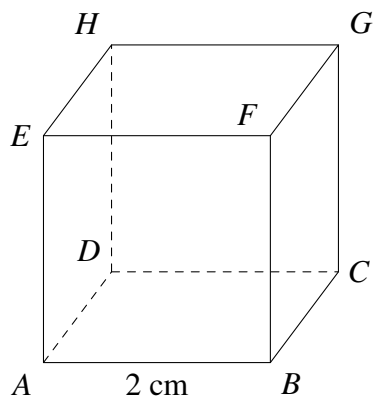
[2]

- 156** The diagram shows a pyramid with a square base $PQRS$.
 The apex V is directly above the centre M of the base.
 The sides of the square base are 6 cm and the vertical height VM is $3\sqrt{2}$ cm.



- (a) Calculate the length PR .
 [2]
- (b) Calculate the length PM .
 [1]
- (c) Calculate the angle between the line VP and the plane $PQRS$.
 [2]

- The diagram shows a cube with base $ABCD$ and top $EFGH$. The side length of the cube is 2 cm.



- (a) Calculate the length FH .

..... [2]

- (b) Calculate the length DF .

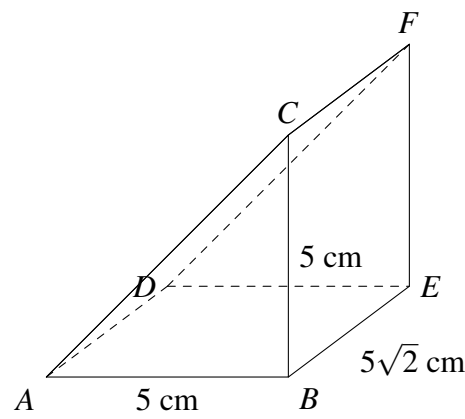
..... [2]

- (c) Show that the cosine of the angle between the line DF and the plane $EFGH$ is

$$\sqrt{\frac{2}{3}}$$

[2]

- 158** The diagram shows a right triangular prism.
 The base $ABED$ is a rectangle.
 The triangles ABC and DEF are right-angled at B and E .
 C is vertically above B and F is vertically above E .
 $AB = 5$ cm, $BC = 5$ cm and $AD = 5\sqrt{2}$ cm.



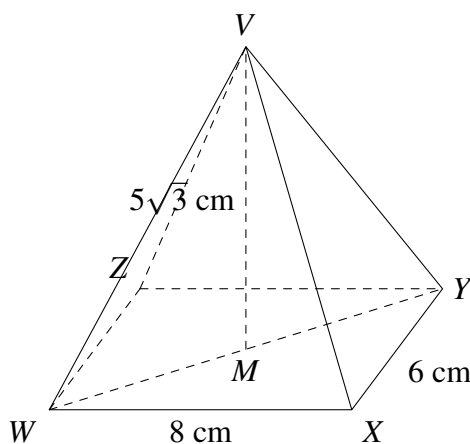
- (a) Calculate the length BD .

..... [2]
- (b) Calculate the length CD .

..... [2]
- (c) Calculate the angle between the line CD and the plane $ABED$.

..... [2]

- 159** The diagram shows a pyramid with a rectangular base $WXYZ$.
The apex V is directly above the centre M of the base.
 $WX = 8$ cm, $XY = 6$ cm and the vertical height $VM = 5\sqrt{3}$ cm.



- (a) Calculate the length WY .

..... [2]

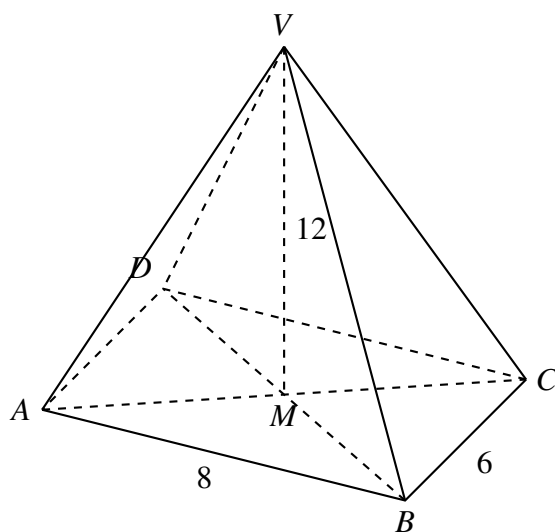
- (b) Calculate the length MY .

..... [1]

- (c) Calculate the angle between the line VY and the plane $WXYZ$.

..... [2]

- 160** The diagram shows a right pyramid $VABCD$ with a rectangular base $ABCD$.
 M is the centre of the base $ABCD$.
 $AB = 8$ cm, $BC = 6$ cm and the vertical height $VM = 12$ cm.



- (a) Calculate the length of AM .

..... [2]

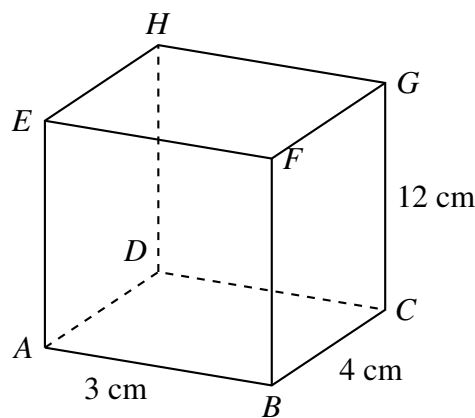
- (b) Calculate the length of VA .

..... [2]

- (c) Show that the cosine of the angle between VA and the plane $ABCD$ is $\frac{5}{13}$.

[2]

161 The diagram shows a cuboid $ABCDEFGH$.



$AB = 3\text{ cm}$, $BC = 4\text{ cm}$ and $CG = 12\text{ cm}$.

(a) Calculate the exact length of AC .

..... [2]

(b) Calculate the exact length of AG .

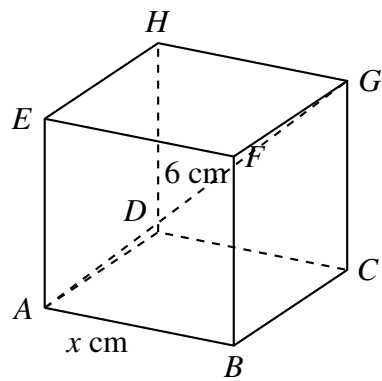
..... [2]

(c) Show that the sine of the angle between the line AG and the plane $ABCD$ is given by the fraction:

$$\frac{12}{13}$$

[2]

162 The diagram shows a cube $ABCDEFGH$ with side length x cm.



The internal diagonal AG has a length of 6 cm.
Write down an expression for AC^2 in terms of x .

..... [1]

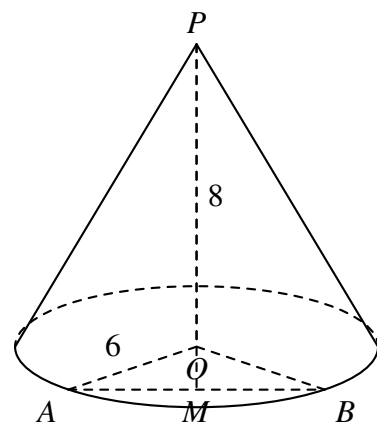
(a) Show that the exact value of x is $2\sqrt{3}$.

[2]

(b) Find the exact value of the sine of the angle between the diagonal AG and the plane $ABCD$.
 $\sin(\theta)$

..... [2]

- 163** A right circular cone has vertex P and the centre of its circular base is O . Points A and B lie on the circumference of the base such that $\angle AOB = 90^\circ$. The radius of the base is 6 and the vertical height PO is 8. The point M is the midpoint of the chord AB .



- (a)

Calculate the exact length of the chord AB .

..... [2]
- (b)

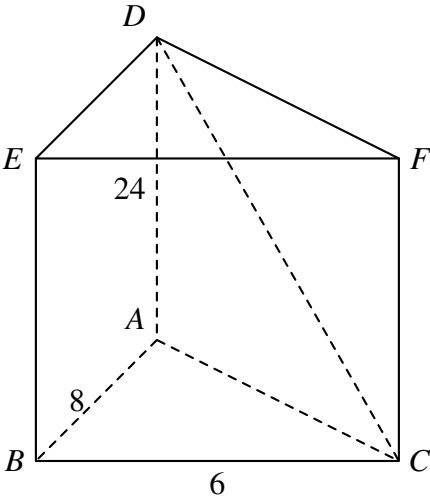
Show that the length OM is $3\sqrt{2}$.

[2]
- (c)

Find the exact value of $\tan(\angle PMO)$, which represents the true angle between the plane PAB and the circular base.

..... [2]

- 164** A solid right prism has vertices A, B, C, D, E, F . The horizontal base $\triangle ABC$ is a right-angled triangle at B . The exact lengths are $AB = 8$, $BC = 6$, and the vertical height of the prism $AD = 24$.



- (a) Calculate the length AC .

..... [2]
- (b) Show that the length of the internal diagonal DC is 26.

[2]
- (c) By calculating the exact square of the length DB , show that $\triangle DBC$ is right-angled at B .

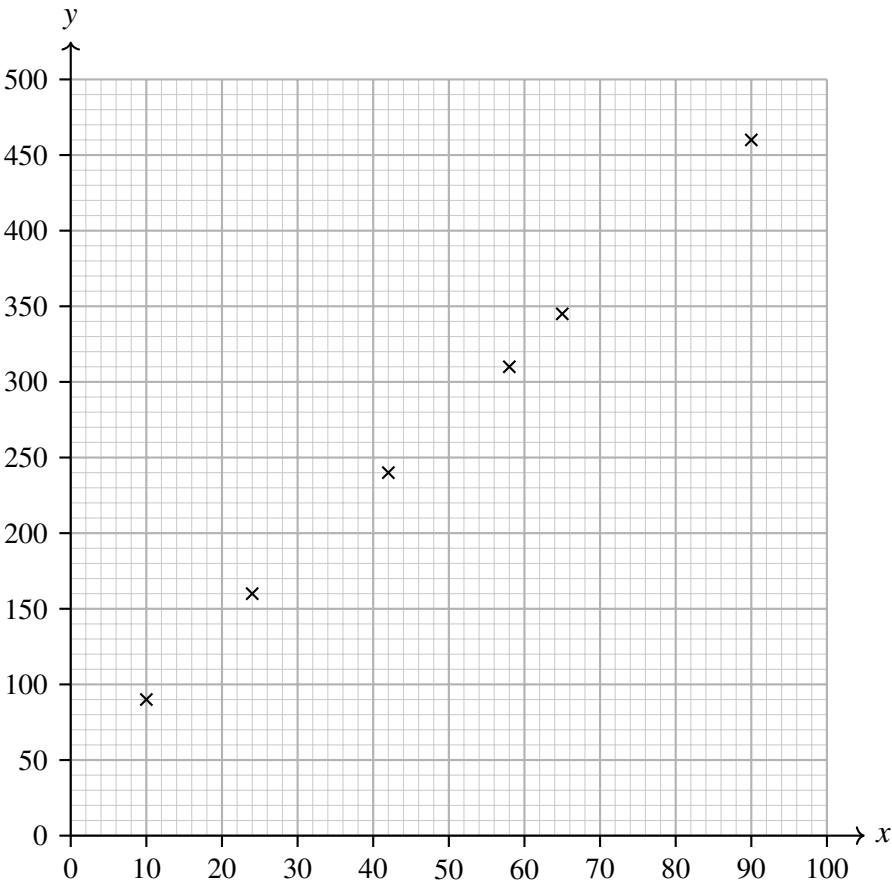
[3]
- (d) Find the exact value of $\cos(\angle DCB)$.

..... [1]

165 A local business analyses its daily advertising budget, x GBP, and the corresponding daily revenue, y GBP, over 8 days. The data is documented in the table below.

Advertising budget (x GBP)	10	24	35	42	58	65	74	90
Daily revenue (y GBP)	90	160	205	240	310	345	380	460

- (a) On the grid, complete the scatter diagram by plotting the remaining two points. The first six points have been plotted for you.



[2]

- (b) State the type of correlation shown in the scatter diagram.

..... [1]

- (c) Use your graphic display calculator to find the equation of the linear regression line, giving

your answer in the form $y = ax + b$.

..... [2]

- (d) (i)** Use the regression equation to predict the daily revenue when the advertising budget is 120 GBP.

..... [2]

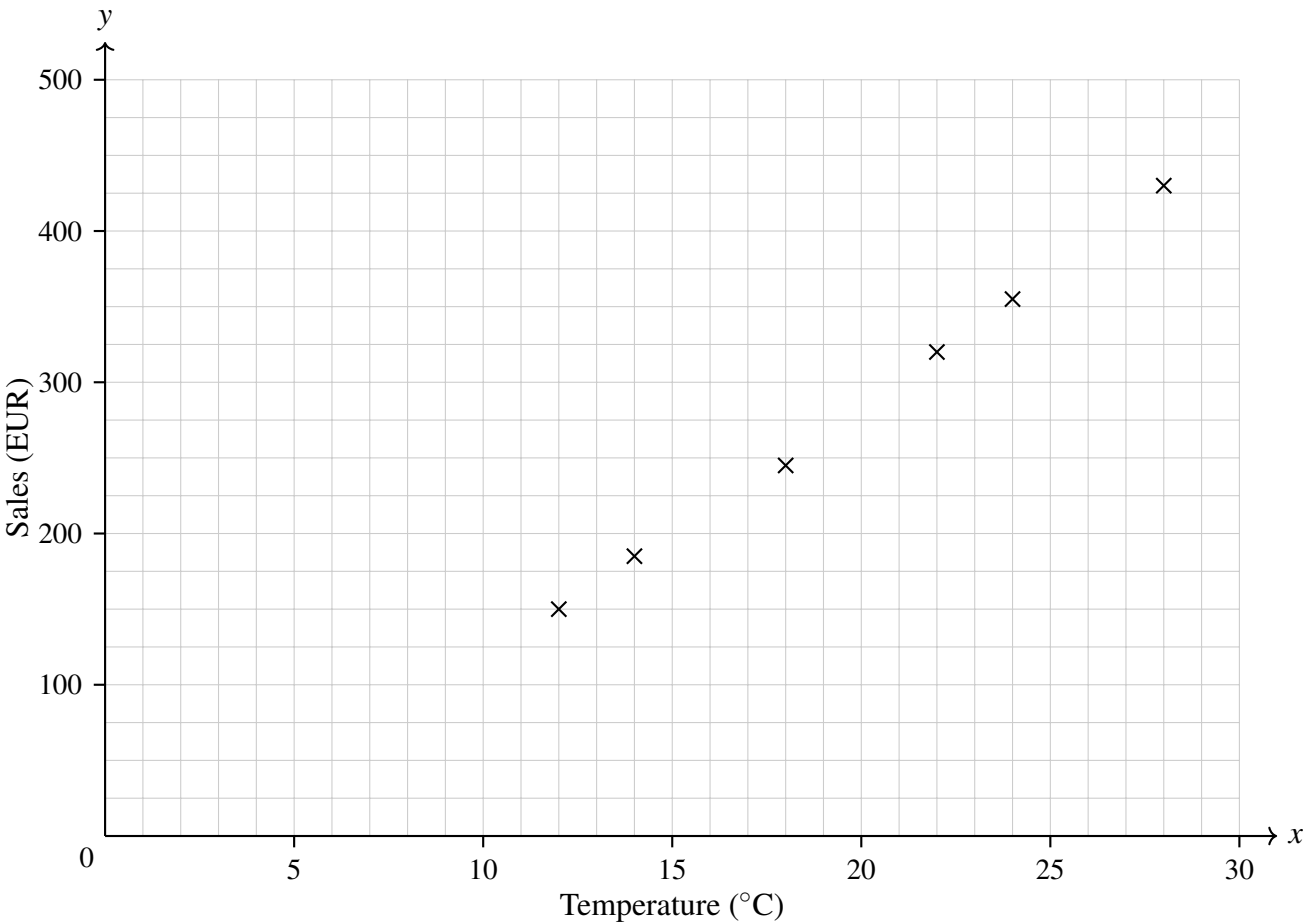
- (ii)** Explain why this prediction might not be reliable.

..... [1]

166 An ice cream vendor records the midday temperature, x in $^{\circ}\text{C}$, and their daily sales, y in EUR, over a period of 8 days. The results are shown in the table.

Temperature (x $^{\circ}\text{C}$)	12	14	16	18	22	24	26	28
Sales (y EUR)	150	185	210	245	320	355	390	430

- (a) Complete the scatter diagram by plotting the missing points for $x = 16$ and $x = 26$. Mark your points using small crosses ($\times$).



[2]

- (b) State the type of correlation shown in the scatter diagram.

..... [1]

- (c) Use your graphic display calculator to find the exact equation of the linear regression line in

the form $y = mx + c$.

..... [2]

- (d) Use your equation to estimate the sales on a day when the temperature is 20°C .

..... [2]

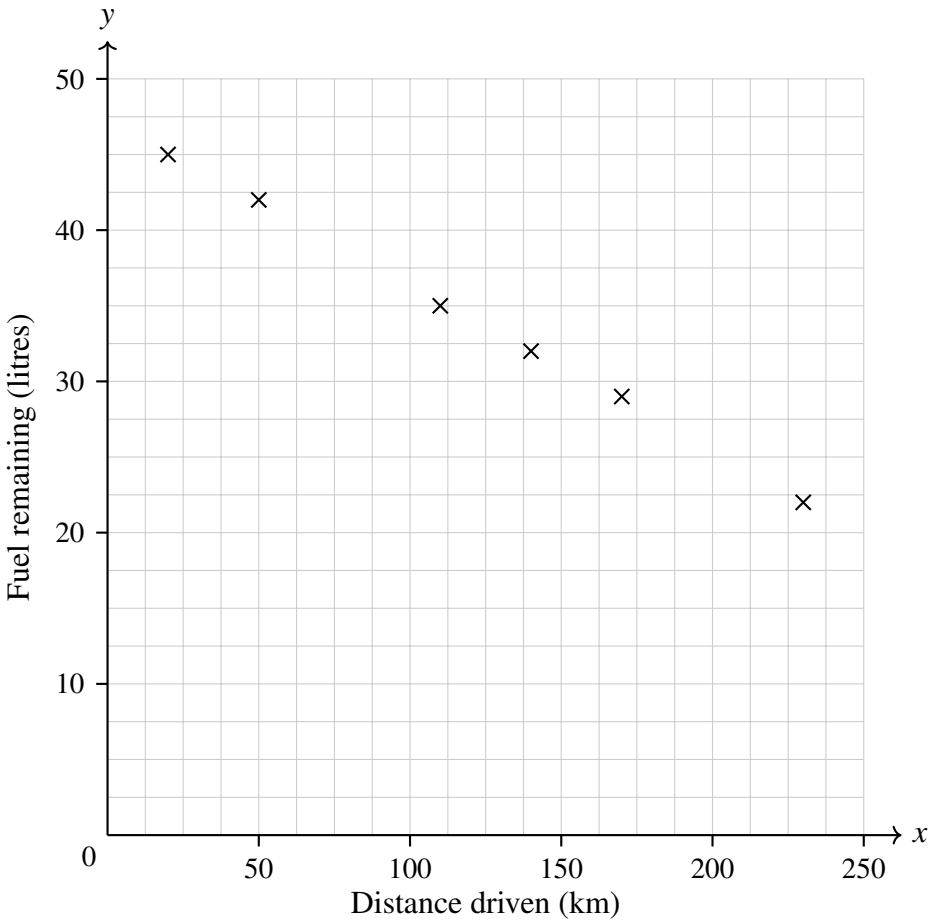
- (e) Is your estimate in part (d) reliable? Give a reason for your answer.

..... [1]

167 The table shows the distance driven, x km, and the amount of fuel remaining in the tank, y litres, for a delivery van.

Distance driven (x km)	20	50	80	110	140	170	200	230
Fuel remaining (y litres)	45	42	38	35	32	29	25	22

- (a) Plot the missing data points for $x = 80$ and $x = 200$ on the grid. Mark all points using small crosses ($\times$).



[2]

- (b) Describe the correlation between distance driven and fuel remaining.

..... [1]

- (c) Use your graphic display calculator to find the exact equation of the linear regression line in

the form $y = mx + c$. Give your parameters to 3 significant figures.

..... [2]

- (d) Predict the amount of fuel remaining after driving 300 km.

..... [2]

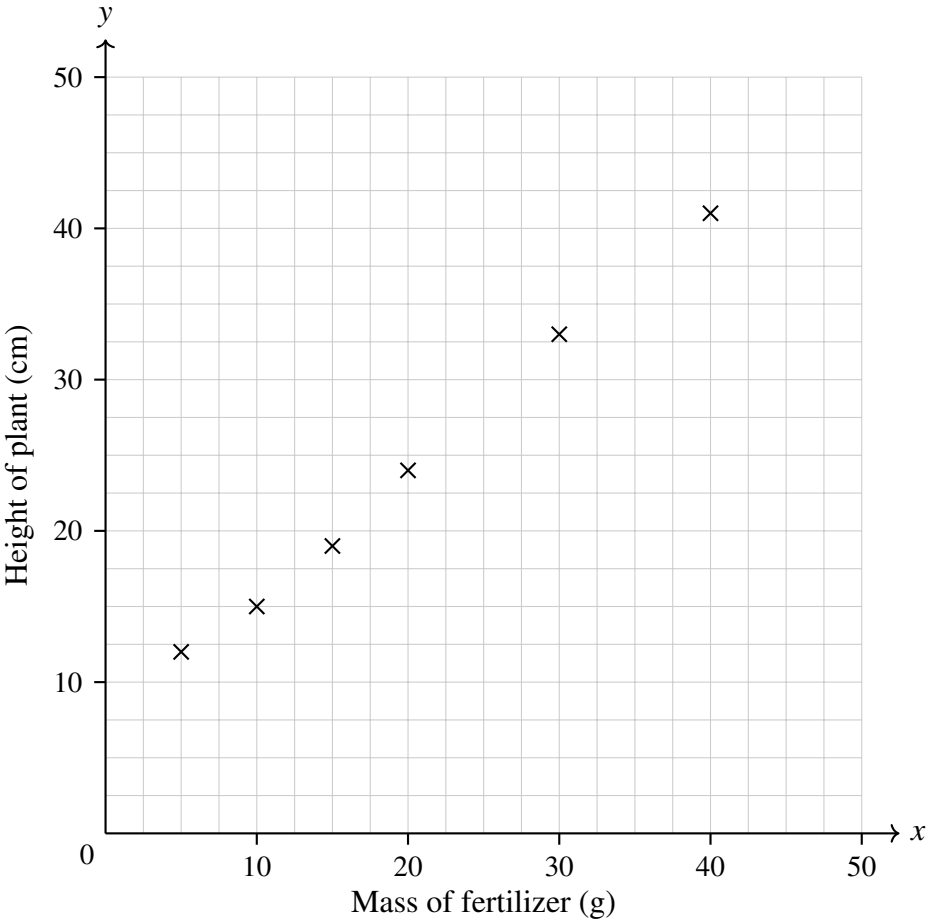
- (e) State whether the prediction in part (d) is an example of interpolation or extrapolation. Explain your reasoning.

..... [1]

168 A botanist tests a new fertilizer on 8 plants. The table records the mass of fertilizer applied, x grams, and the resulting height of the plant, y cm, after two months.

Mass of fertilizer (x g)	5	10	15	20	25	30	35	40
Height of plant (y cm)	12	15	19	24	28	33	36	41

- (a) Two points are missing from the scatter diagram below. Plot the points corresponding to $x = 25$ and $x = 35$, marking them with small crosses ($\times$).



[2]

- (b) State the correlation shown by your scatter graph.

..... [1]

- (c) Use your graphic display calculator to determine the equation of the line of best fit in the

form $y = mx + c$.

..... [2]

- (d) Using your equation, calculate the expected height of a plant that receives 18 g of fertilizer.

..... [2]

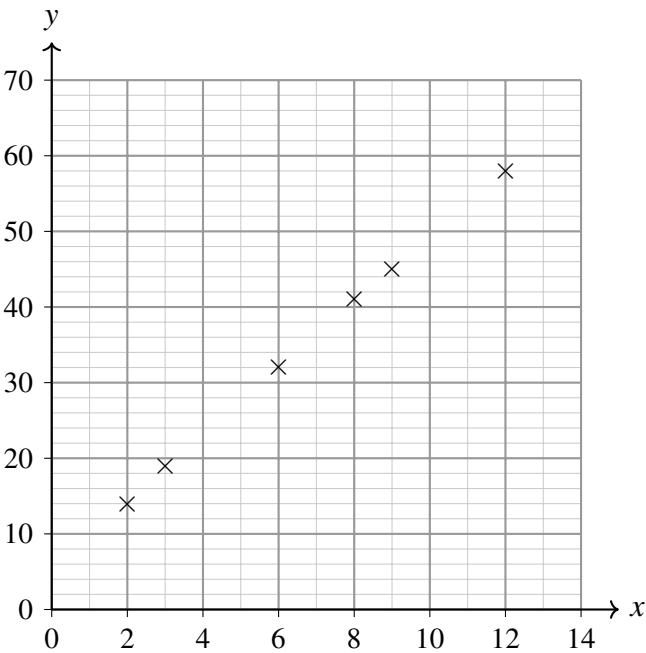
- (e) Give a reason why your prediction in part (d) is considered reliable.

..... [1]

169 A farmer records the mass of fertiliser applied (x kg) and the corresponding yield of tomatoes (y kg) from 8 identical plots of land.

Mass of fertiliser (x kg)	2	3	5	6	8	9	11	12
Yield of tomatoes (y kg)	14	19	27	32	41	45	52	58

- (a) (a) Six of these points have been plotted on the scatter diagram below. Complete the diagram by plotting the remaining two points as small crosses ($\times$).



[2]

- (b) (b) Describe the correlation between the mass of fertiliser applied and the yield of tomatoes.

..... [1]

- (c) (c) Utilise a graphic display calculator to determine the equation of the linear regression line, $y = mx + c$.

..... [2]

- (d) (d) Use your regression equation to estimate the yield of tomatoes when 10 kg of fertiliser is

applied.

..... [2]

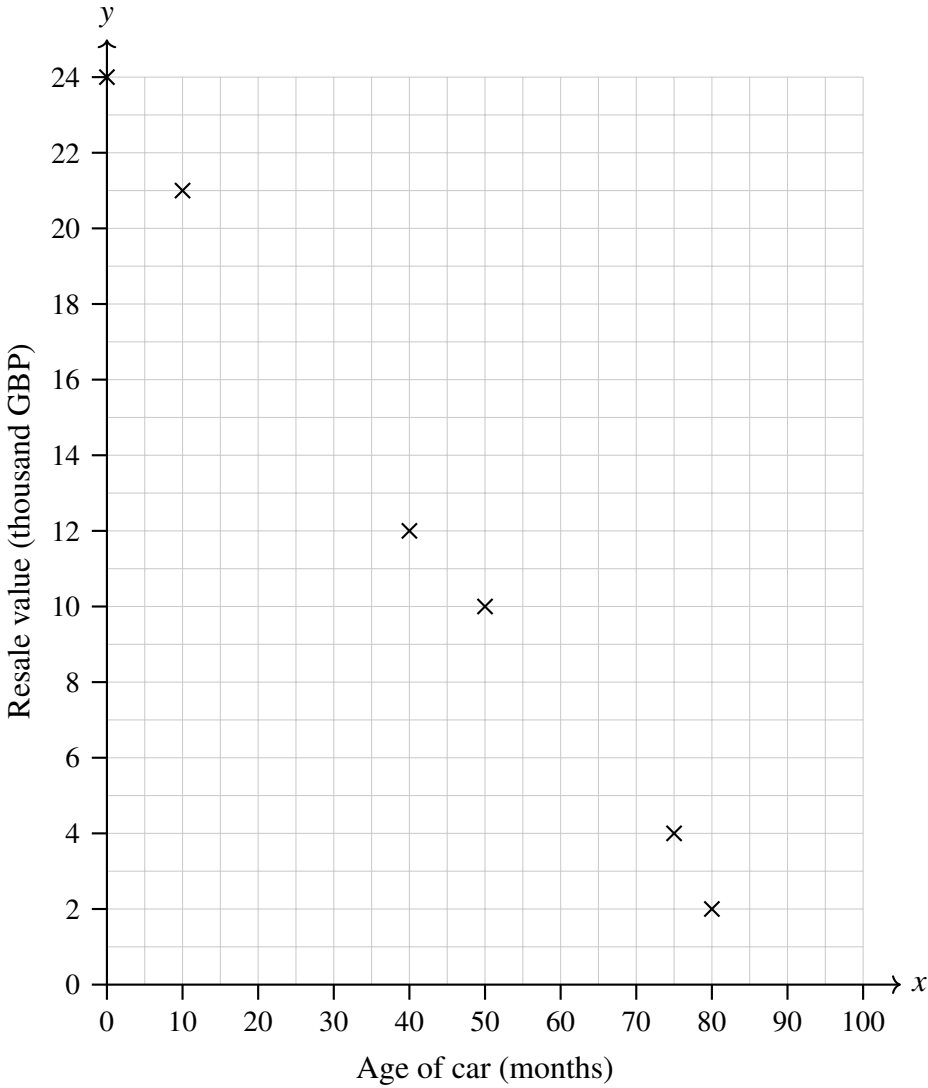
- (e) (e) The farmer wants to use the equation to predict the yield if 16 kg of fertiliser is applied. Explain why this prediction might not be reliable.

..... [1]

170 A dealership analyses the relationship between the age of a specific car model in months, x , and its resale value in thousands of GBP, y . The data for 8 cars is given below.

Age of car (x months)	0	10	25	40	50	60	75	80
Resale value (y thousand GBP)	24	21	17	12	10	8	4	2

- (a) Six points have already been plotted as small crosses ($\times$). Complete the scatter diagram by plotting the remaining two points as small crosses.



[2]

- (b) State the type of correlation shown in the scatter diagram.

..... [1]

- (c) Find the exact equation of the linear regression line in the form $y = ax + b$. Give your param-

eters to 3 significant figures.

..... [2]

- (d) Estimate the resale value of a car that is 35 months old.

..... [1]

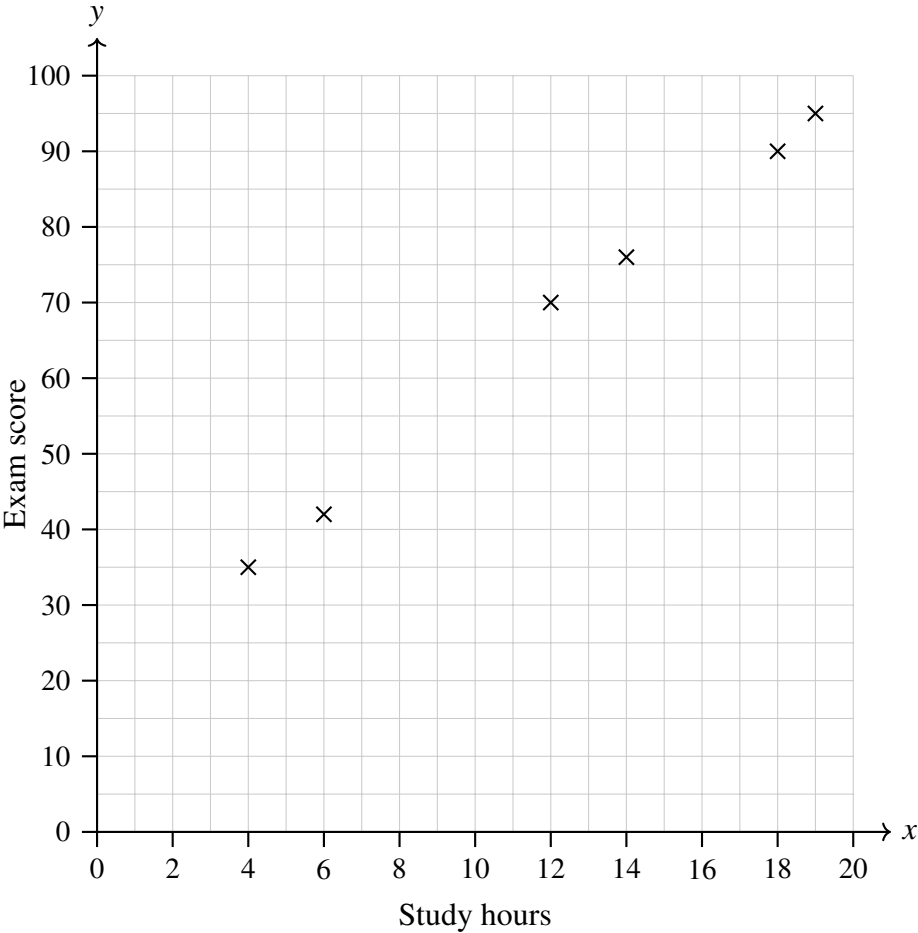
- (e) Explain why it might not be sensible to use this regression equation to estimate the resale value of a car that is 120 months old.

..... [1]

171 A researcher is investigating the relationship between the number of hours students spend studying independently over a month, x , and their final examination score, y . The data for 8 students is recorded in the table below.

Study hours (x)	4	6	9	12	14	15	18	19
Exam score (y)	35	42	58	70	76	82	90	95

- (a) Some points have been plotted as small crosses (×) on the grid for you. Complete the scatter diagram by plotting the remaining two points as small crosses.



[2]

- (b) What type of correlation does the scatter diagram show?

..... [1]

- (c) Use a graphic display calculator to find the exact equation of the linear regression line, giving your answer in the form $y = ax + b$. Give your values of a and b correct to 3 significant

figures.

..... [2]

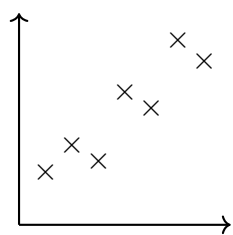
- (d) Use your equation to estimate the exam score of a student who studied for 10 hours.

..... [1]

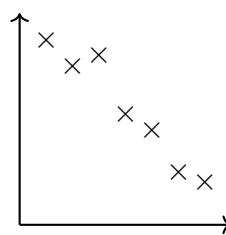
- (e) A student suggests using the regression line to predict the score of someone who studied for 30 hours. State whether this prediction is valid and give a mathematical reason for your answer.

..... [1]

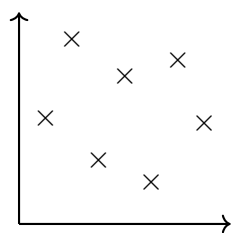
- 172** A student investigates the relationship between various pairs of variables. Four scatter diagrams, labelled A, B, C, and D, show different types of correlation.



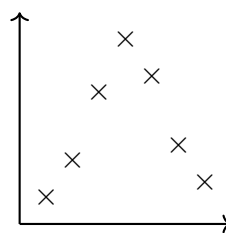
A



B



C



D

(a) State which diagram illustrates zero correlation.

..... [1]

(b) State which diagram illustrates a strong negative correlation.

..... [1]

(c) Describe the relationship between the variables plotted in Diagram A.

..... [1]

173 (a) A large shipping container has a volume of 12 m^3 .

(i) Convert this volume into cubic centimetres (cm^3).

..... [1]

(ii) Convert this volume into litres.

..... [1]

(b) A block of metal has a mass of 4.5 kg . Convert this mass into grams (g).

..... [1]

174 (a) A rectangular floor tile has an area of 600 cm^2 .

(i) Convert this area into square millimetres (mm^2).

..... [1]

(ii) Convert this area into square metres (m^2).

..... [1]

(b) A bottle contains 350 ml of water. Convert this capacity into litres (l).

..... [1]

175 (a) A field has an area of 0.05 km^2 .

(i) Convert this area into square metres (m^2).

..... [1]

(ii) Convert 150 mm^2 into square centimetres (cm^2).

..... [1]

(b) A storage box has a volume of $80\,000 \text{ cm}^3$. Convert this volume into cubic metres (m^3).

..... [1]

- 176** (a) A structural beam has a length of 2.8 m.
(i) Convert this length into millimetres (mm).

..... [1]

- (ii) Convert 250 g of flour into kilograms (kg).

..... [1]

- (b) A glass contains 0.45 l of juice. Convert this capacity into cubic centimetres (cm³).

..... [1]

-
- 177** A solid gold statue has a volume of 350 cm³.

- (a) Convert the volume of the statue into cubic millimetres (mm³).

..... [1]

- (b) The density of gold is 19.3 g/cm³. Calculate the mass of the statue in kilograms.

..... [2]

- (c) A replica of the statue is made from a polymer where every linear dimension is scaled down by a factor of $\frac{1}{5}$. Calculate the volume of the replica in cm³.

..... [2]

178 A large cylindrical water tank has a cross-sectional area of 2.5 m^2 . Water is poured into the tank at a constant rate of 150 ml per second.

(a) Express the flow rate of the water in litres per minute.

..... [2]

(b) Show that the height of the water in the tank increases by 3.6 mm each hour.

[3]

179 A cubic container has a total capacity of 0.027 m^3 .

(a) Find the capacity of the container in litres.

..... [1]

(b) Find the length of one edge of the container, giving your answer in centimetres.

..... [2]

(c) The container is filled with oil. A small pipe drains the oil at a rate of 45 ml per second.

Calculate the time, in minutes, required to completely empty the full container.

..... [2]

180 A square agricultural test plot has a side length of 0.04 km.

- (a) Convert the side length of the test plot into millimetres (mm). Give your answer in standard scientific notation $A \times 10^n$.

..... [2]

- (b) Calculate the total area of the agricultural test plot in square metres (m^2).

..... [2]

- (c) A liquid nutrient treatment is applied uniformly over the plot at a rate of 150 ml per square metre. Calculate the total volume of nutrient treatment used over the entire plot, giving your final answer in litres (l).

..... [2]

181 A hypothetical dataset maps the depth of an underwater probe, x metres, against the received signal strength index, y units. A line of best fit is calculated from this survey and given by the linear regression equation:

$$y = 92.5 - 0.38x$$

- (a) State the type of correlation that this equation implies between the depth of the probe and the received signal strength index.

..... [1]

- (b) Use the equation to calculate the predicted signal strength index when the probe is at a depth of 45.0 metres.

..... [1]

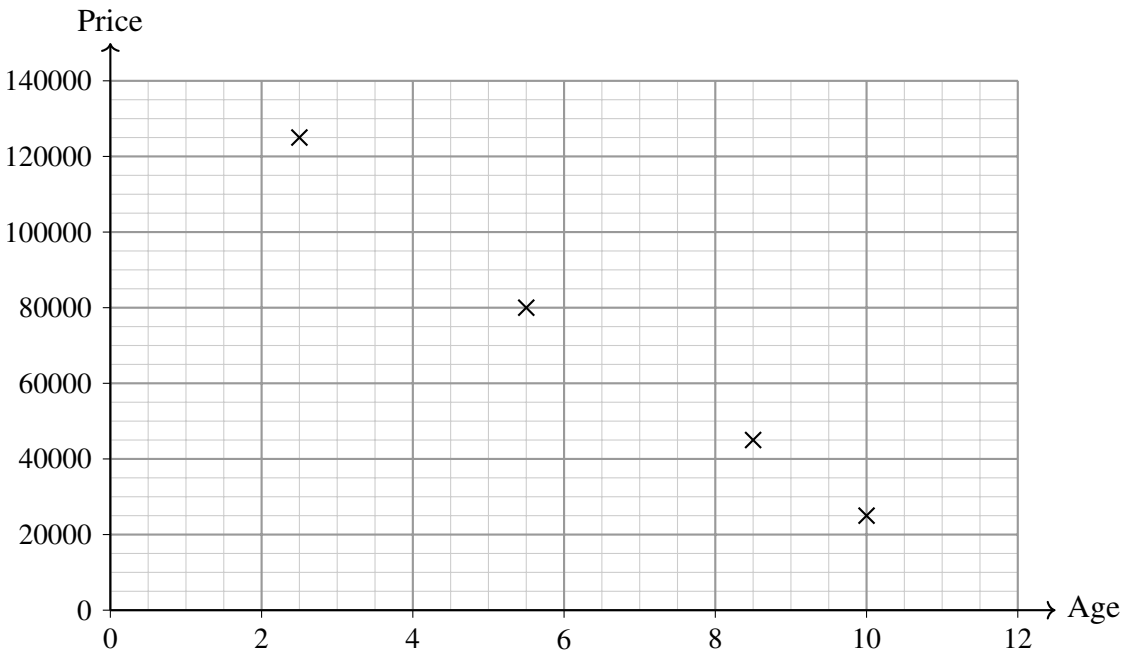
- (c) Find the predicted depth of the probe when the signal strength index drops to 64.0 units.

..... [2]

182 A dealership records the age of used vehicles, x years, and their selling price, y in HKD.

Age (x)	2.5	4.0	5.5	7.0	8.5	10.0
Price (y)	125000	102500	80000	62500	45000	25000

(a) Complete the scatter diagram by plotting the remaining two points.



[2]

(b) State the type of correlation.

..... [1]

(c) Find the equation of the linear regression line in the form $y = mx + c$.

..... [2]

(d) Predict the selling price of a vehicle that is 6.5 years old.

..... [2]

183 A logistics company tracks the number of warehouse staff on duty, x , and the time taken to process a standard shipment, y minutes. Data was collected for shifts where the number of staff members was strictly between 8 and 22. The linear regression equation for the time taken is calculated as:

$$y = 342.7 - 12.85x$$

(a) State the type of correlation between the number of staff and the processing time.

..... [1]

(b) The mean number of staff on duty during the observed shifts is 16.

(i) Calculate the mean time taken to process a standard shipment.

..... [2]

(ii) Predict the processing time when 20 staff members are on duty.

..... [2]

(c) A shipment is processed in 150 minutes. Estimate the number of staff on duty, giving your answer to the nearest whole number.

..... [3]

(d) State whether it is appropriate to use this equation to predict the processing time if 35 staff members are on duty. Give a clear reason for your answer.

..... [1]

184 A company purchases a piece of industrial machinery for 50000 EUR. The value of the machinery depreciates exponentially at a rate of 20% each year.

(a) Show that the value of the machinery after 2 years is 32000 EUR.

[2]

(b) Calculate the value of the machinery after 3 years.

..... [2]

185 The population of a small village is tracked using an exponential growth model. The initial population of the village is 1200. The population increases at a constant rate of 5% each year.

(a) Find the population of the village after 1 year.

..... [1]

(b) Calculate the population of the village after 2 years.

..... [2]

186 A delivery fleet vehicle was bought for 16000 AUD. The vehicle's value loses value exponentially at a rate of 50% per year.

(a) Calculate the value of the vehicle after 3 years.

..... [2]

(b) Determine the total value lost by the vehicle over these 3 years.

..... [1]

187 A herd of rare deer in a nature reserve has an initial population of 480. The population increases exponentially at a rate of 6.4% per year.

(a) Calculate the expected population of the herd after 5 years. Give your answer to the nearest whole number.

..... [2]

(b) Calculate the number of complete years it will take for the population of the herd to exceed 1000.

..... [3]

188 A manufacturing company purchases a heavy industrial machine for 75000 EUR. The value of the machine depreciates exponentially at a constant rate of 12.5% per year.

(a) Calculate the value of the machine after 4 years. Give your answer to the nearest EUR.

..... [2]

(b) Find the total percentage depreciation in the value of the machine over these 4 years.

..... [2]

189 An investor purchases a vintage car as an asset for 42000 AUD. The value of the asset increases exponentially at a fixed rate of 5.8% per year.

(a) Calculate the value of the vintage car after 6 years. Give your answer correct to the nearest 100 AUD.

..... [2]

(b) Calculate the number of complete years required for the original investment value to increase by at least 75% .

..... [3]

190 A tech company tracks the active user base of a new digital platform. The initial group consists of 12500 active users and increases exponentially at a rate of 18.2% per month.

- (a) Calculate the number of active users after 8 months. Give your answer correct to 3 significant figures.

..... [2]

- (b) Find the number of complete months it takes for the user base to reach 5 times its initial size.

..... [3]

-
- 191** A regional wildlife reserve tracks the population of an endangered species of wetland birds. At the start of 2018, a comprehensive census records the population at 1420 birds. Due to habitat restoration and conservation efforts, the population increases at a continuous compound rate of 4.3% per year.

- (a) Calculate the expected population of the wetland birds at the start of 2026. Give your answer to the nearest whole number.

..... [2]

- (b) Find the number of complete years it will take for the initial population of 1420 birds to double.

..... [3]

-
- 192** A manufacturing logistics company purchases a fleet of heavy-duty delivery vehicles for a total initial asset value of 485000 EUR. The financial accounting team applies a reducing-balance depre-

ciation model where the total value of the fleet decreases at a constant rate of 12.6% per annum.

- (a) Calculate the value of the fleet of vehicles at the end of 6 years. Give your answer correct to 3 significant figures.

..... [2]

- (b) Show that the value of the fleet decreases to less than half of its initial purchase value during the 6th year.

[3]

-
- 193** An marine biology study monitors the population of a specific invasive fish species in a closed bay ecosystem. Biologists estimate the population at 8500 fish at the beginning of a monitoring phase. Due to resources becoming scarce, the population decreases at a constant rate of 6.8% each month.

- (a) Calculate the estimated population of the invasive fish species after 1.5 years. Give your answer to the nearest hundred fish.

..... [3]

- (b) Determine the number of months required for the population to decrease from 8500 to below 2500 fish.

..... [3]

194 The population of Town A is currently 45000 and is decreasing at a rate of 1.8% per year. The population of Town B is currently 32000 and is increasing at a rate of 2.4% per year.

- (a) Calculate the population of Town A after exactly 5 years, giving your answer correct to 3 significant figures.

..... [2]

- (b) Show that after n years, the condition for Town B’s population to exceed Town A’s population simplifies to the inequality:

$$\left(\frac{1.024}{0.982}\right)^n > \frac{45}{32}$$

[2]

- (c) Hence calculate the minimum integer value of n for which Town B has a greater population than Town A.

..... [3]

195 A manufacturing company purchases a specialized robotic machine for 125000 USD. The machine depreciates in value by 15% per year for the first 3 years. It then depreciates by $r\%$ per year for every subsequent year.

- (a) Calculate the value of the machine after exactly 3 years.

..... [2]

- (b) After exactly 8 years, the total value of the machine is 41500 USD. Calculate the value of r , correct to 3 significant figures.

..... [3]

- (c) A different robotic machine depreciates at a constant rate of 12% per year. Its value today is 28160 USD. Calculate its original value exactly 6 years ago.

..... [3]

- 196** A nature reserve introduces a population of 850 deer. The population increases by 8% per year for the first 5 years. A specific disease then causes the population to decrease by $k\%$ per year subsequently.

- (a) Calculate the number of deer at the end of the first 5 years, giving your answer to the nearest whole number.

..... [2]

- (b) Ten years after the initial introduction, the population has fallen to 720 deer. Calculate the value of k , correct to 3 significant figures.

..... [3]

- (c) Assuming the disease continues at the exact same exponential rate, calculate the total number of years from the initial introduction until the population first drops below 500 deer.

..... [4]

197 A rare gemstone is bought for 15000 GBP. Its value increases exponentially at a constant percentage rate. After 4 years, its value is exactly 18500 GBP.

- (a) Calculate the annual percentage increase in the value of the gemstone, correct to 3 significant figures.

..... [3]

- (b) A gold watch is bought at the same time for V GBP. Its value decreases by 3.5% per year. After exactly 7 years, the value of the gold watch is exactly half the value of the gemstone. Show that $V = 13900$ GBP, correct to 3 significant figures.

[3]

- (c) Calculate the minimum number of whole years it takes from the time of purchase for the gemstone's value to be more than three times the gold watch's value.

..... [4]

198 (a) Simplify completely.

$$\sqrt{32}$$

..... [1]

- (b) Rationalise the denominator and simplify.

$$\frac{12}{\sqrt{6}}$$

..... [2]

- 199 (a) Simplify the expression.

$$\sqrt{75} + \sqrt{12}$$

..... [2]

- (b) Rationalise the denominator.

$$\frac{10}{\sqrt{5}}$$

..... [2]

- 200 (a) Simplify.

$$\sqrt{80}$$

..... [1]

- (b) Show that the following expression simplifies to $5\sqrt{3}$.

$$\frac{15}{\sqrt{3}}$$

[2]

- 201 (a) Simplify.

$$\sqrt{28} - \sqrt{7}$$

..... [2]

- (b) Rationalise the denominator.

$$\frac{8}{\sqrt{2}}$$

..... [2]

- 202 (a) (a) Simplify fully.

$$3\sqrt{48} - \sqrt{12}$$

..... [2]

(b) (b) Rationalise the denominator and simplify fully.

$$\frac{5 + \sqrt{3}}{2 + \sqrt{3}}$$

..... [3]

203 **(a)** (a) Simplify fully.

$$\sqrt{98} - \sqrt{50}$$

..... [2]

(b) (b) Rationalise the denominator.

$$\frac{8}{3 + \sqrt{5}}$$

..... [3]

204 (a) (a) Expand and simplify.

$$(2\sqrt{5} - 3)(\sqrt{5} + 2)$$

..... [2]

(b) (b) Show that the expression reduces to $3\sqrt{2}$.

$$\frac{12}{\sqrt{8}}$$

[2]

205 (a) (a) Simplify fully.

$$\sqrt{125} + 2\sqrt{20}$$

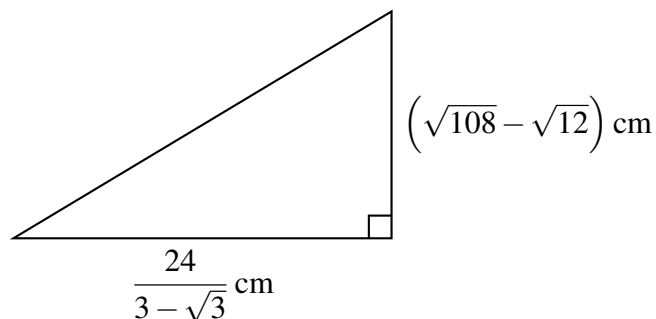
..... [2]

(b) (b) Rationalise the denominator.

$$\frac{10}{\sqrt{7} - \sqrt{2}}$$

..... [3]

206 A right-angled triangle has dimensions as follows:



(a) Show that the perpendicular height of the triangle simplifies exactly to $4\sqrt{3}$ cm.

[2]

(b) Rationalise the denominator of the base and simplify your answer fully.

$$\frac{24}{3 - \sqrt{3}}$$

..... [3]

(c) Calculate the exact area of the triangle in the form $a + b\sqrt{3}$.

..... [3]

207 (a) Solve the equation for x . Express your final answer in the form $p + q\sqrt{2}$.

$$x\sqrt{18} - 4 = x\sqrt{2} + \sqrt{50}$$

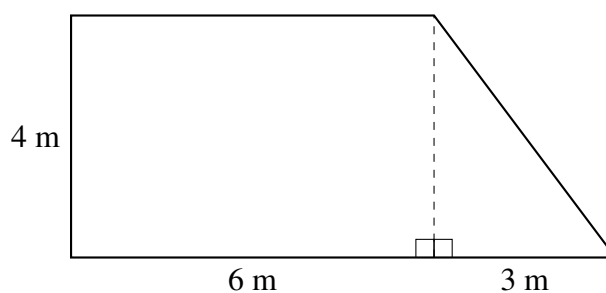
..... [4]

- (b) Rationalise the denominator and simplify fully.

$$\frac{7 - \sqrt{2}}{3 + 2\sqrt{2}}$$

..... [3]

-
- 208** A room's floor is in the shape of a rectangle joined to a right-angled triangle.



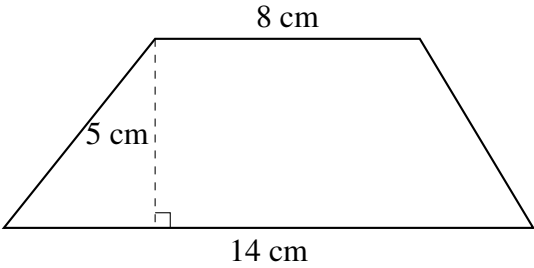
- (a) Calculate the total area of the floor.

..... [2]

- (b) The cost to carpet the floor is 15 GBP per square metre. Calculate the total cost to carpet the floor.

..... [2]

209 The diagram shows a trapezium.



(a) Calculate the area of the trapezium.

..... [2]

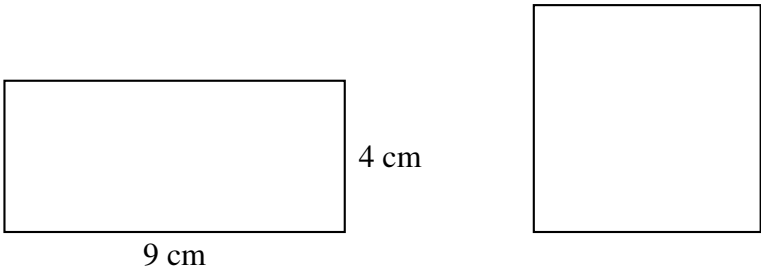
(b) A distinct triangle has the same area as the trapezium. The triangle has a base of 11 cm and a perpendicular height of h cm.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Show that the height, h , of the triangle is 10 cm.

[2]

210 A rectangle has length 9 cm and width 4 cm.
A square has the exact same area as the rectangle.



(a) Calculate the area of the rectangle.

..... [1]

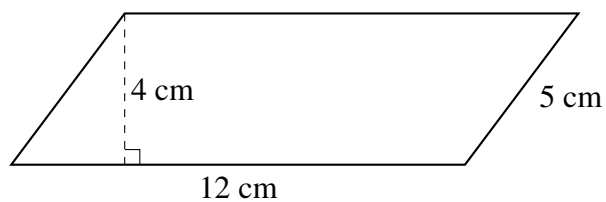
(b) Find the side length of the square.

..... [2]

(c) Calculate the perimeter of the square.

..... [1]

211 The diagram shows a parallelogram.



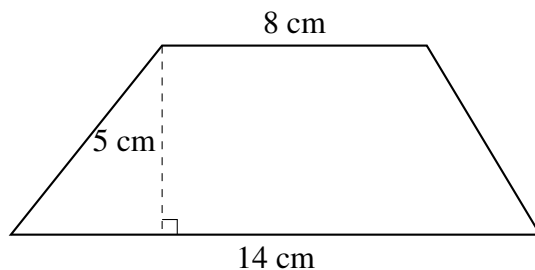
(a) Calculate the perimeter of the parallelogram.

..... [1]

(b) Calculate the area of the parallelogram.

..... [1]

-
- 212** The diagram shows a trapezium.



- (a) Calculate the area of the trapezium.

..... [2]

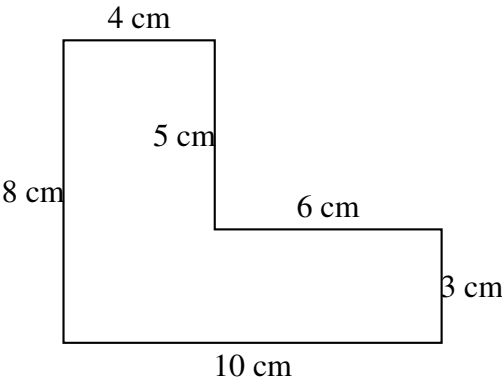
- (b) A distinct triangle has the same area as the trapezium. The triangle has a base of 11 cm and a perpendicular height of h cm.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Show that the height, h , of the triangle is 10 cm.

[2]

213 The diagram shows a composite L-shaped polygon.
All internal angles are right angles.



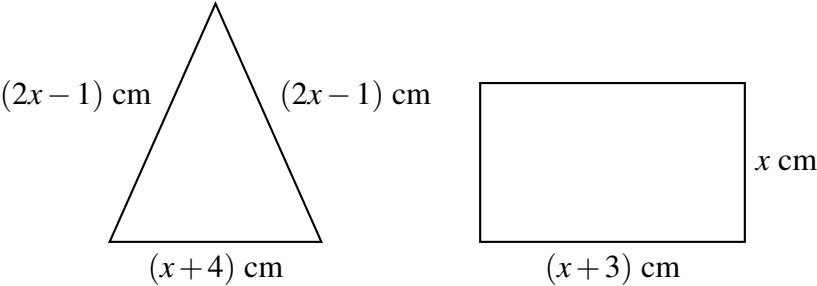
(a) Calculate the perimeter of the polygon.

..... [1]

(b) Calculate the total area of the polygon.

..... [3]

214 An isosceles triangle and a rectangle are shown below.



The perimeter of the triangle is equal to the perimeter of the rectangle.

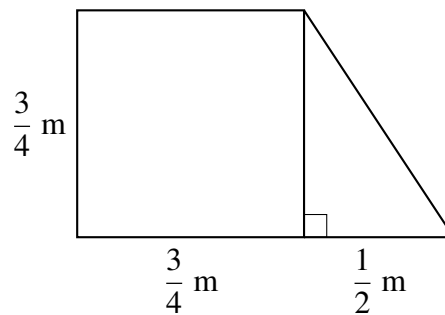
(a) Show that the equation simplifies to find $x = 4$.

[3]

(b) Calculate the area of the rectangle.

..... [2]

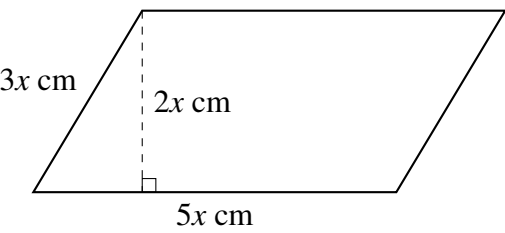
215 The diagram shows a composite shape formed by joining a square and a right-angled triangle.



Calculate the exact total area of the composite shape. Give your answer as a single fraction in its simplest form.

..... [4]

216 A parallelogram is shown below.



The area of the parallelogram is 40 cm^2 .

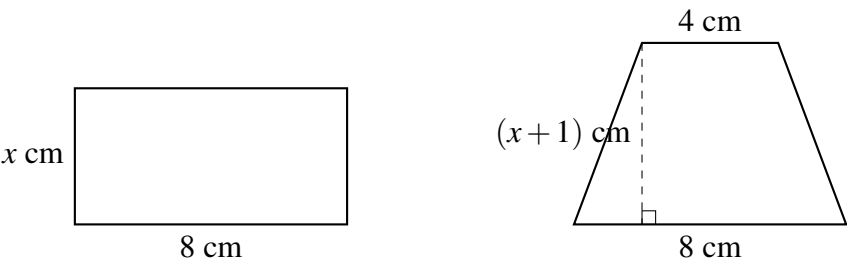
(a) Calculate the value of x , given that $x > 0$.

..... [3]

(b) Calculate the perimeter of the parallelogram.

..... [2]

217 A rectangle and a trapezium are shown below.



The area of the rectangle is equal to the area of the trapezium.

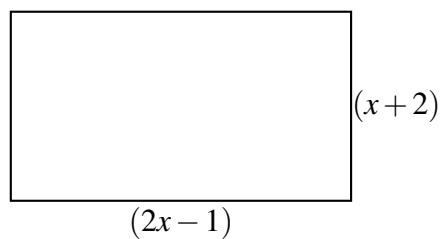
(a) Find the value of x .

..... [3]

(b) Calculate the perimeter of the rectangle.

..... [1]

218 A rectangle is shown below. All dimensions are in centimetres.



The exact area of the rectangle is 18 cm^2 .

(a) Show that the area equation simplifies to:

$$2x^2 + 3x - 20 = 0$$

[2]

(b) Solve the equation to find the value of x .

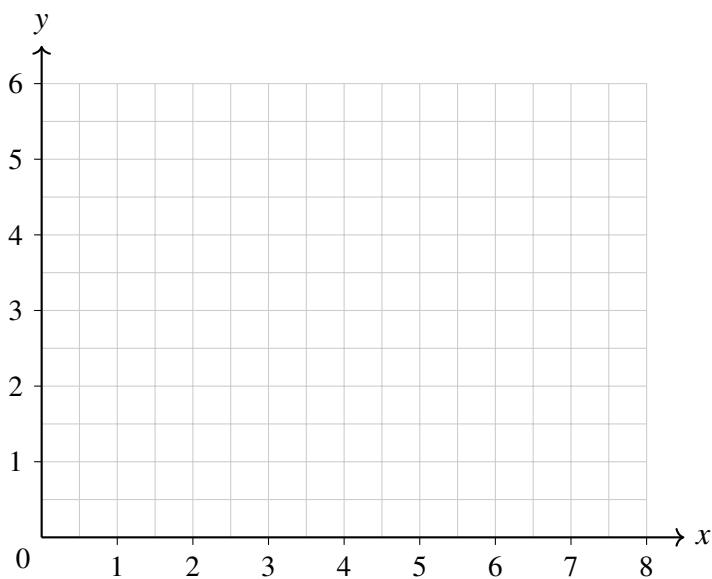
..... [3]

(c) Calculate the exact perimeter of the rectangle.

..... [2]

219 The coordinates of the vertices of a polygon are $A(1, 1)$, $B(5, 1)$, $C(7, 5)$, and $D(3, 5)$.

(a) On the grid provided, draw the parallelogram $ABCD$.



[2]

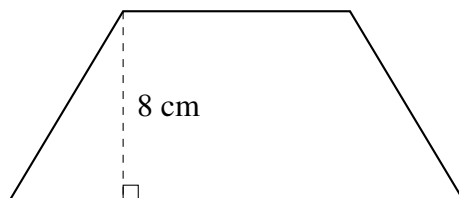
(b) Calculate the exact area of the parallelogram $ABCD$.

..... [2]

- (c) Calculate the exact perimeter of the parallelogram $ABCD$. Give your answer in the form $p + q\sqrt{r}$ where p, q , and r are integers.

..... [3]

- 220** The parallel sides of a trapezium are in the ratio 3 : 5.



The perpendicular height of the trapezium is 8 cm.

The area of the trapezium is 96 cm^2 .

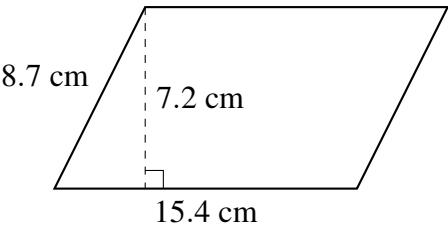
- (a) Calculate the lengths of the two parallel sides.

..... [3]

- (b) The trapezium is isosceles. Calculate the exact perimeter of the trapezium. Give your answer in the form $p + q\sqrt{r}$ where p, q , and r are integers.

..... [4]

- 221** The diagram shows a parallelogram with a base of 15.4 cm and a slant edge of 8.7 cm. The perpendicular height of the parallelogram is 7.2 cm.



A student is analysing the geometric properties of the shape.

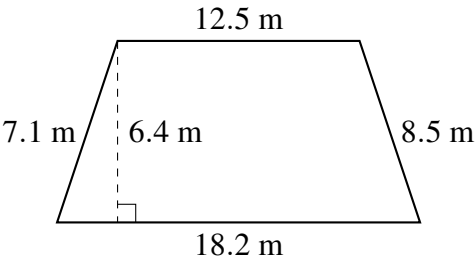
- (a) Calculate the perimeter of the parallelogram.

..... [2]

- (b) Calculate the area of the parallelogram.

..... [2]

- 222** The diagram shows a trapezium. The parallel sides have lengths of 18.2 m and 12.5 m. The perpendicular distance between the parallel sides is 6.4 m. The lengths of the remaining two sides are 7.1 m and 8.5 m.



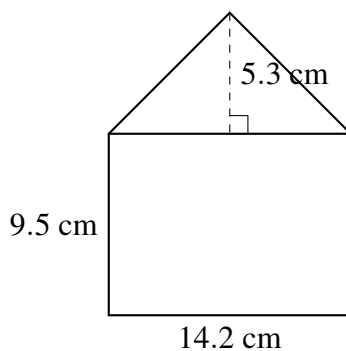
- (a) Calculate the perimeter of the trapezium.

..... [2]

- (b) Calculate the area of the trapezium.

..... [2]

-
- 223** The diagram shows a compound shape consisting of a rectangle and a right-angled triangle. The rectangle has a length of 14.2 cm and a width of 9.5 cm. The triangle has a perpendicular height of 5.3 cm.



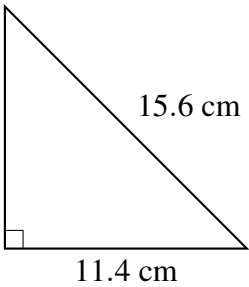
- (a) Calculate the total area of the compound shape.

..... [3]

- (b) A protective liquid coating is applied to the surface of the shape. The cost of the coating is 0.15 USD per square centimetre. Calculate the total cost of the coating.

..... [2]

- 224** The diagram shows a right-angled triangle. The base length is 11.4 cm and the hypotenuse is 15.6 cm.



- (a) Calculate the perpendicular height of the right-angled triangle.

..... [3]

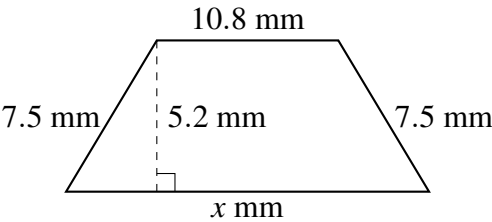
- (b) The formula for the area of a triangle is:

$$Area = \frac{1}{2}bh$$

Calculate the area of the triangle.

..... [2]

- 225** The diagram shows a symmetrical trapezium with parallel sides of length 10.8 mm and x mm. The perpendicular height is 5.2 mm and the two slant sides are each 7.5 mm in length.



The total area of the trapezium is 85.8 mm². The formula for the area of a trapezium is:

$$A = \frac{1}{2}(a + b)h$$

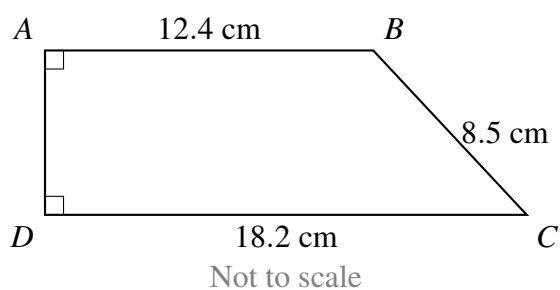
- (a) Show that the value of x is 22.2.

[2]

- (b) Calculate the perimeter of the trapezium.

..... [2]

- 226** The diagram shows a trapezium $ABCD$. AB is parallel to DC . Angle $ADC = 90^\circ$.
 $AB = 12.4$ cm, $DC = 18.2$ cm, and $BC = 8.5$ cm.



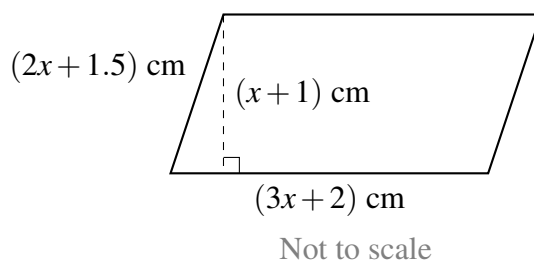
- (a) Calculate the perimeter of the trapezium $ABCD$.

..... [3]

- (b) Calculate the area of the trapezium $ABCD$.

..... [2]

-
- 227** The diagram shows a parallelogram. The base is $(3x + 2)$ cm and the perpendicular height is $(x + 1)$ cm. The slant edge is $(2x + 1.5)$ cm. The area of the parallelogram is 56 cm^2 .



- (a)** Show that x satisfies the equation

$$3x^2 + 5x - 54 = 0$$

[2]

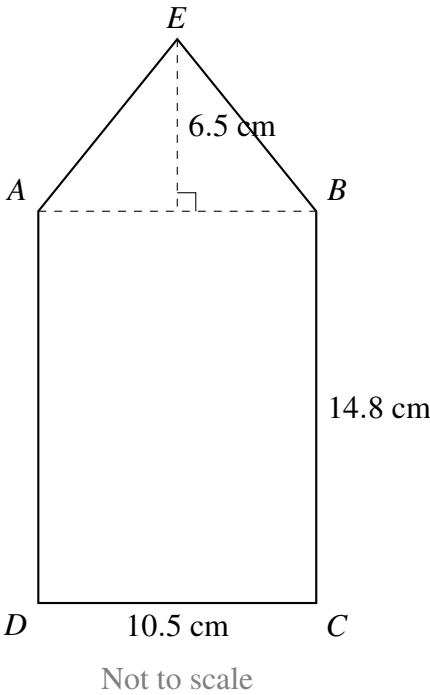
- (b)** Solve the equation $3x^2 + 5x - 54 = 0$ to find the value of x . Give your answer correct to 3 significant figures.

..... [3]

- (c)** Calculate the perimeter of the parallelogram.

..... [2]

- 228** A metal plate is formed from a rectangle $ABCD$ and an isosceles triangle ABE . $AB = 10.5$ cm and $BC = 14.8$ cm. The perpendicular height of the triangle ABE from the base AB is 6.5 cm.



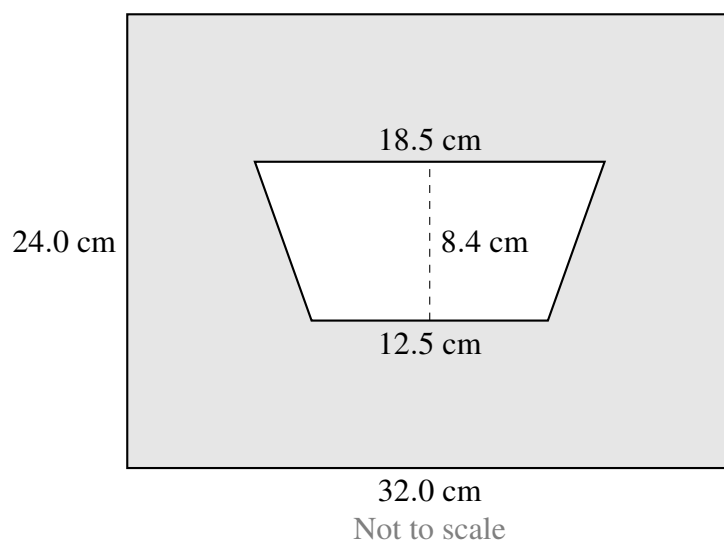
(a) Calculate the total area of the metal plate.

..... [3]

(b) Calculate the perimeter of the metal plate.

..... [3]

- 229** A rectangular piece of card measures 32.0 cm by 24.0 cm. A hole in the shape of an isosceles trapezium is cut from the centre of the card, leaving the shaded region shown in the diagram. The parallel sides of the trapezium are 12.5 cm and 18.5 cm. The perpendicular height of the trapezium is 8.4 cm.



- (a) Calculate the area of the shaded region.

..... [3]

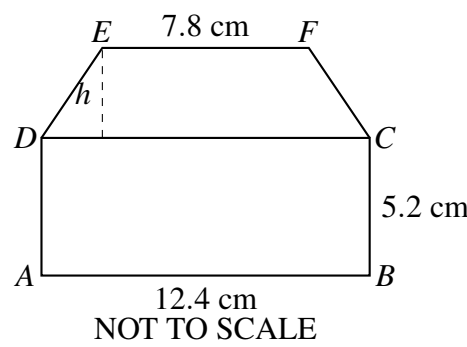
- (b) Calculate the perimeter of the cut-out isosceles trapezium.

..... [3]

- (c) The shaded region is painted. The cost of the paint is 0.045 EUR per square centimetre. Calculate the total cost of painting the shaded region.

..... [2]

- 230** A cross-section of a solid is formed by a rectangle $ABCD$ and an isosceles trapezium $DCEF$ placed adjacent to DC .
 The width of the rectangle is $AB = 12.4$ cm and the height is $BC = 5.2$ cm.
 The top edge of the trapezium EF is parallel to DC and $EF = 7.8$ cm.
 The total area of the composite cross-section is 98.82 cm².



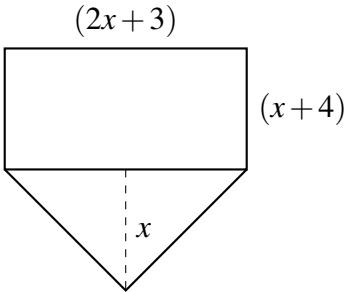
- (a) Show that the perpendicular height h of the trapezium $DCEF$ is 3.4 cm.

[2]

- (b) Calculate the total perimeter of the cross-section.

..... [4]

- 231** A polygon consists of a rectangle and an isosceles triangle joined at their common base. The rectangle has side lengths of $(2x + 3)$ cm and $(x + 4)$ cm. The attached isosceles triangle has a base of $(2x + 3)$ cm and a perpendicular height of x cm. The total area of the entire polygon is 110 cm^2 .



NOT TO SCALE

- (a) Show that $6x^2 + 25x - 196 = 0$.

[3]

- (b) Solve the equation $6x^2 + 25x - 196 = 0$ to find the value of x .

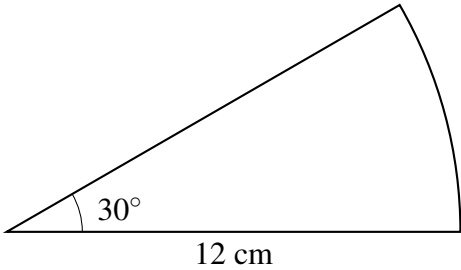
..... [2]

- (c) Hence, calculate the total perimeter of the polygon.

..... [4]

232 A minor sector of a circle has a radius of 12 cm and an angle of 30° .
 The area, A , of a sector is determined by the formula:

$$A = \frac{\theta}{360} \times \pi r^2$$



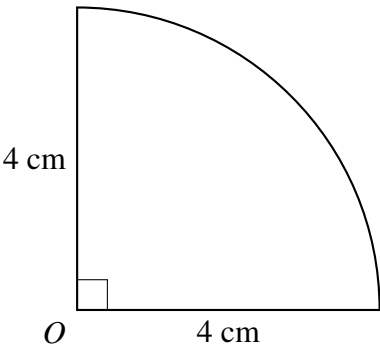
(a) Calculate the area of the sector.

..... [2]

(b) Find the perimeter of the sector in the form $a + b\pi$, where a and b are integers.

..... [3]

233 The diagram shows a sector of a circle with centre O and radius 4 cm.
 The angle of the sector is 90° .



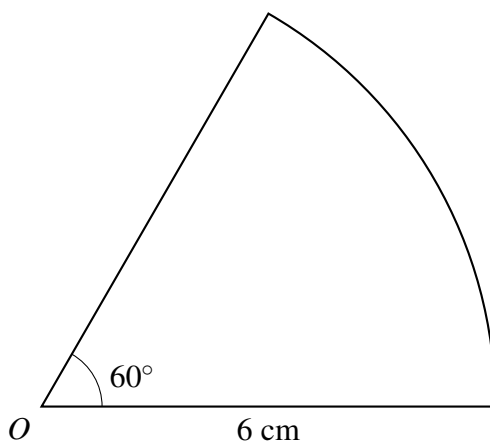
(a) Calculate the exact area of the sector.

..... [2]

(b) Calculate the exact perimeter of the sector.

..... [3]

-
- 234** The diagram shows a sector of a circle with centre O and radius 6 cm.
The sector angle is 60° .



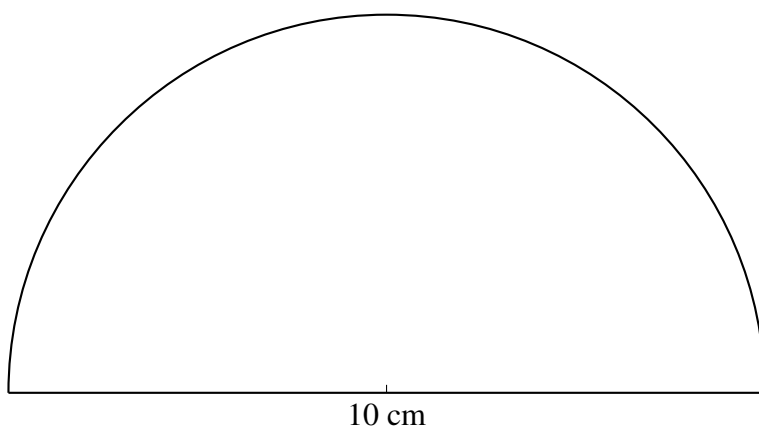
(a) Show that the exact arc length of the minor sector is 2π cm.

[2]

- (b) Calculate the exact area of the major sector in terms of π .

..... [2]

-
- 235 The diagram shows a semicircle with a straight diameter of 10 cm.



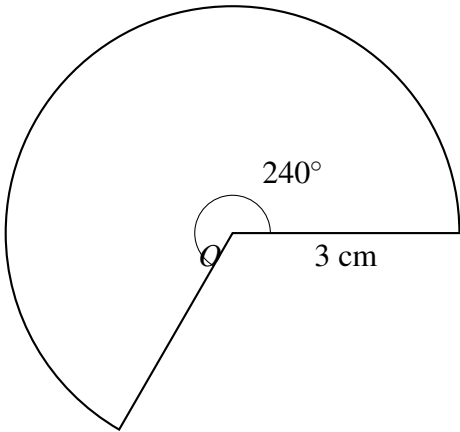
- (a) Calculate the exact area of the semicircle.

..... [2]

- (b) Calculate the total perimeter of the semicircle, leaving your answer in terms of π .

..... [2]

- 236** A ceremonial badge is shaped as a major sector of a circle with centre O and radius 3 cm. The reflex angle of the major sector is 240° .



- (a) Show that the exact area of the badge is $6\pi\text{ cm}^2$.

- (b) The badge is coated in liquid silver. The silver coating process costs 5 GBP per $\pi\text{ cm}^2$.
- (i) Calculate the total cost to coat the badge.

[2]

..... [2]